Attitudinal and spatial drivers of policy targeting for nearshore management in Oregon: Integrating latent class choice models with two-dimensional spatial GAMs

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Abstract

Environmental valuation studies often assume that preferences are constant across space, yet support for conservation policies can vary systematically by where people live, driven by the likelihood of belonging to heterogeneous preference segments detected. Using a discrete choice experiment of Oregon residents, we evaluate preferences for changes to the state Marine Reserve System (MRS) and test whether heterogeneity takes the form of distinct segments, geographic clustering, and distance decay. In a novel approach combining latent class model (LCM), Global Moran's I and two-dimensional spatial generalized additive models (GAMs), we diagnose and map spatial structure in both segment-membership probabilities and attribute utilities. Five preference segments were identified, with most respondents preferring policy changes relative to the status quo. Spatial tests show that several segment-membership probabilities and MRS attributes are geographically clustered and present a distance-decay pattern toward the coast. Contrary to a simple local-versus-global goods prediction, the results reveal a pronounced divide: coastal areas place greater value on blue carbon management and are more likely to belong to pro-environmental segments, whereas inland regions place greater weight on ocean job creation and, in some areas, on the status quo. Furthermore, spatial diagnostics show that incorporating spatial smoothing sharpens inference, improving the LCM's individual estimates of average preferences. These findings support targeting policy communication by region rather than using a single statewide message, and clarify that the welfare gains from marine-reserve enhancements are unevenly distributed across Oregon.

Keywords: non-market valuation; management of marine reserves; environmental policy; latent class model; generalized additive model; spatial heterogeneity of preferences; discrete choice experiment

1 Introduction

Environmental policies to protect and restore ecosystem services have become increasingly important for addressing climate change and other environmental pressures, as they can generate benefits such as carbon sequestration, improved air and water quality, and biodiversity conservation. Knowing how people value and prefer ecosystem services is crucial for designing effective policies to protect and enhance them. Because many of these benefits are not traded in conventional markets, stated-preference valuation is often needed to quantify welfare changes and to support cost–benefit assessments. For results to inform policy properly, valuation needs to generate realistic estimates and explicitly examine the relevant components and drivers behind the preferences being modeled.

Stated-preference research indicates that valuation estimates are influenced by psychological and cultural drivers such as attitudes, worldviews, and environmental beliefs, as

well as by where people live and what opportunities they face nearby (Hoyos et al., 2009; Bateman, et al., 2006; De Valck and Rolfe, 2018; Eusse-Villa et al., 2021). One way to summarize heterogeneity in acceptability for environmental policies is to identify distinct preference segments and the drivers that characterize each segment, using a Latent Class Model (LCM) (Boxal and Adamowicz, 2022). Identifying these preference segments is important to characterize the overall pattern of the population preferences, giving us how population can be subdivided by distinct opinions toward the environment. Therefore, it can help policymakers improve policy communication to the population, understand what people prioritize when formulating policy, and assess the acceptability of policy components. In addition, preferences for ecosystem services have often been shown to be spatially structured rather than being spatially uniform (Johnston and Ramachandran, 2014; Czajkowski et al., 2017; Schaafsma et al., 2012; De Valck and Rolfe, 2018; Glenk et al., 2020). The proximity to the environmental good in question affects how individuals evaluate, accept, and prefer it. In many cases, the literature finds a pattern in which valuation decreases as individuals live farther from the environmental good being valued, which is commonly described as distance decay (Bateman et al., 2006; Schaafsma et al., 2013; Johnston et al., 2019; Ruiz-Frau et al., 2019). Location-dependent preferences can also arise when respondents living near one another tend to reveal similar values (Johnston and Ramachandran, 2014; Czajkowski et al., 2017; Badura et al., 2019). In this case, preferences for an environmental good can cluster/concentrate in specific regions, implying that some locations may exhibit higher or lower values than others systematically, even after accounting for distance to the environmental good itself. Because there is evidence that spatial dependence in stated-preference data often appears as areas of relatively high or low willingness to pay (WTP) rather than as purely random noise (Johnston and Ramachandran, 2014; Czajkowski et al., 2017; Campbell et al., 2008; De Valck and Rolfe, 2018), it is reasonable to investigate whether some areas can also be interpreted as geographically distinct preference segments. These two factors – distance and spatial clustering - when incorporated can improve the realism of estimates, strengthen the interpretation of preference heterogeneity, and provide a clearer understanding of the evaluation pattern in relation to the environmental good. Also, it can sharpen inference about where support is stronger or weaker, which is directly relevant for policy communication and political feasibility, in the sense that knowing where preferences cluster allows policymakers to design communication strategies that are sensitive to regional differences rather than relying on a single statewide message. From an implementation perspective, identifying where willingness to support (and pay for) improvements is stronger can also inform how policies are prioritized, financed, and sequenced, and when implementation is spatially flexible, it can inform where policy may be located.

Beyond purely spatial frictions, behavioral components, such as general environmental attitudes, place-based identity beliefs (a relational component), and broader world-

views/values influence preferences creating a plausible channel linking where people live with what they believe and what they prefer. In the discrete-choice tradition, these psychological drivers can enter the utility model either as observed covariates (e.g., attitudinal scales) or as latent constructs that shape taste heterogeneity, rather than being absorbed only as "unobserved heterogeneity" in the error term or random coefficients (McFadden, 1974; Ben-Akiva et al., 2002, Hoyos et et al., 2015; Faccioli et al., 2020; Morey et al., 2006; Aldrich et al., 2007; Meldrum et al., 2015). Attitudes refer to relatively general evaluative orientations toward the environment, while place identity captures relational/attachment-based beliefs tied to a specific place and its landscape, and these are conceptually distinct even if they move together empirically in some settings (Gatersleben et al., 2014; Faccioli et al. 2020, Pascual et al., 2023). A location may shape attitudes/identity through local experience and residence history, and conversely, people may also sort across space in ways correlated with their values and lifestyles (so the relationship is not necessarily "causal" in one direction). In practice, location might be linked with attitudes/worldviews and these psychological components are correlated with preference structure. Thus, spatial patterns in preferences may reflect both economic mechanisms (distance/substitutes) and "idiosyncratic" drivers outside simple distance stories (Johnston and Ramachandran, 2014; Glenk et al., 2019; Faccioli et al., 2020). More broadly, spatial patterns may be linked to "why" preferences differ in the sense that location is often correlated with beliefs and attitudes because of social context, lived experience, and residential sorting (Tiebout, 1956; Banzhaf and Walsh, 2008). This motivates a framework where spatial structure is not treated as noise, but as part of the empirical story connecting proximity, exposure, and underlying attitudes to observed stated preferences.

There are distinct forms of environmental services that generate use values and non-use values (Krutilla, 1967; Loomis and White, 1996). Use values arise when individuals benefit from direct (e.g., recreation, harvesting, or fisheries spillover benefits) or indirect use of the resource (e.g., regulating services such as carbon sequestration, pollination and habitat provision), whereas non-use values arise even without use (e.g., existence and bequest motives). Ecosystem services also differ in the spatial scale over which benefits accrue - from predominantly local or regional benefits tied to proximity and opportunity sets to broadly distributed national/global benefits (as climate regulation) - which provides an initial guidance about when preferences may be more geographically structured (Hein et al., 2006; Fisher et al., 2009). These goods also differ in the extent to which conservation actions can be sited and scaled across alternative candidate locations, which conditions how precisely policies can be targeted to places where ecological benefits and/or human benefits are highest (Ando et al., 1998; Polasky et al., 2008; Cjaizkovski et al, 2020). Moreover, for site-specific environmental improvements - particularly those yielding use values - the marginal value in a given location depends on the availability of

close spatial substitutes in the sense that when similar sites or opportunities are abundant nearby, willingness to pay for changes at any one site tends to be lower, whereas scarce substitutes imply higher marginal value (Parsons and Hauber, 1998; Schaafsma and Brouwer, 2020). Thus, for policies that are more spatially mobile, implementation can be targeted not only to locations where ecological benefits are higher, but also to settings where fewer close substitutes imply higher marginal benefits. For example, a policy to restore native vegetation or expand forest conservation can be shifted across alternative land areas, allowing implementation to be targeted toward locations where public benefits are higher. In contrast, a policy to conserve ocean ecosystems through marine reserves has limited spatial mobility because it can only be implemented along the coast and suitable nearshore habitats.

These distinctions between value types and spatial constraints are particularly salient when evaluating marine reserves. Marine reserves are designated areas intended to protect the marine environment, fish populations, related ecological processes and to conduct research and learning. They have also gained climate relevance through blue carbon, which is the conservation and restoration of carbon-sequestering coastal and marine ecosystems, including mangroves, seagrass beds, tidal marshes, and kelp forests (High Level Panel for a Sustainable Ocean Economy. 2023). This nature-based form of carbon capture has been highlighted as a promising and potentially cost-effective approach to climate mitigation, with evidence that the public may be more supportive of nature-based solutions than engineered alternatives (Bertram and Merk, 2020; Nawaz et al., 2023; Merk et al., 2023) and that such interventions can generate co-benefits such as jobs and community improvements (BenDor et al., 2015). In Oregon, a Marine Reserve System (MRS) has been established since 2012, consisting of five sites spanning about 9% of the coastal shoreline (ODFW, 2022). The MRS encompasses both marine protected areas, where some extractive uses are allowed, and marine reserves, where extraction and development are prohibited. In addition, no active management beyond monitoring is allowed in marine reserves.

Introducing blue carbon initiatives within this system together with potential changes in reserve size and co-benefits bundles distinct environmental services that generate both use, non-use, global and local values, differ in substitutability, and are spatially constrained to suitable coastal habitats. These policy changes should, therefore, be evaluated in terms of Oregon residents' preferences and acceptability taking into account the drivers of those outcomes, recognizing that the distinct elements of the bundle may exhibit different spatial patterns of preferences.

This lead us to the following three questions and hypothesis:

Do people living closer to the coast have more pro-environmental attitudes than people who lives further to the coast?

It reflects extensive evidence of distance-decay in environmental valuations, whereby

individuals nearer to the environmental good in question (that generates the ecosystem services) should be more likely to value it higher and sort into conservation-oriented segments than those farther inland, other factors equal (Schaafsma et al, 2013; Johnston et al, 2019; Ruiz-Frau et al., 2019).

H1: All else equal, individuals closer to the coast are more likely to value more pro-environmental changes in MRS and belong to conservation-worldview preference segments, while those farther inlands are more likely to value less the pro-environmental changes and to belong to segments characterized by weak environmental concerns.

The second question is: do preferences reveal spatial clustering - people who live closer are more likely to share the same preferences - of geographically distinct citizen preference segments and attributes acceptability for MRS policy changes?

Consistent with spatial analysis theory, where positive spatial autocorrelation signifies clustering rather than random scatter (Anselin, 2020), and with stated-preference studies that explicitly map spatial variation in preferences on environmental goods (Campbell et al, 2008, Campbell et al., 2009; Johnston Ramachandran, 2014; Czajkowski et al., 2017; Badura et al, 2020; Glenk et al., 2020), we expect preference segments to exhibit geographic clustering across respondent locations and that the probability of belonging to each preference segment varies smoothly over space (Wood, 2008; Leathwick et al. 2006; Elith Leathwick, 2009).

H2: People who live nearby tend to share same preferences than people further away. Residents located closer together are more likely to share the same preference segment and attribute valuation level than residents located farther apart - segment membership and attribute preferences are spatially autocorrelated rather than random.

The final question is: do locally salient, proximity-dependent MRS attributes that is susceptible to changes exhibit stronger spatial structure preferences than a more diffuse/global attribute?

In environmental services, the scope of beneficiaries and the salience of benefits can vary geographically (Hein et al., 2006; Fisher et al., 2009). Stated-preference valuation which accounts for site-based improvements whose benefits depend on proximity, access, and local opportunity sets commonly show distance decay and spatially heterogeneous values, including evidence that distance-decay can differ between use and non-use components (Bateman et al., 2006; Han-ley et al., 2003). We expect that an environmental service that generates a diffuse benefit may exhibit distance-based gradients and spatial dependence that are weaker on average than those for local benefits (De Valck and Rolfe, 2018; Bateman et al., 2006; Dallimer et al., 2014). However, spatial structure may still emerge if environmental attitudes and place-based identities that shape valuation cluster geographically for the attribute evaluated (Faccioli et al., 2020; Iversen and Dugstad, 2024; Bakhtiari et al., 2018).

H3: Preferences for MRS attributes that generate local benefits will show stronger

spatial patterning (distance decay or spatial clustering) than preferences for MRS attributes that generate global benefits.

To address these hypotheses, it is necessary to consider the various analytical frameworks used to identify spatial heterogeneity in the valuation of ecosystem services, which has been analyzed and identified in multiple ways. Different methodological approaches capture this phenomenon through mechanisms such as distance decay, substitutability, spatial clustering, and related spatial structures (Glenk, 2019; DeValcke and Rolfe, 2017). Several studies adopt two-step strategies in which individual-level welfare measures are first estimated and then examined using spatial lag models or spatial autocorrelation tests (Czajkosvky et al., 2016; Budzinski et al., 2017; Lu et al., 2025; Yao et al., 2019). Other contributions incorporate linear or nonlinear functions to represent distance decay and spatial dependence directly, improving the behavioral plausibility of estimates when values vary with proximity or accessibility (Shaafsma et al., 2013; Ferrini and Fezzi, 2012; Rolfe and Windle, 2012; Olsen et al., 2020). A complementary set of studies maps preferences by extrapolating estimated values across space incorporating geostatistical tools (Johnson and Ramacharadan, 2014; Johnston et al., 2015; Eusse-Villa et al., 2025; Danley et al., 2021; Johnston et al., 2019; Czajkosvky et al. 2017; Bateman et al., 2016; Campbell et al., 2008; Campbell et al., 2009; Faccioli et al., 2020; Yao et al., 2019). Additional approaches embed spatial dimensions directly in survey designs (Badura et al.. 2019; Munro et al., 2017; Brouwer et al., 2010, Wallmo and Kosaka, 2017; Faccioli et al., 2020), define distance bands to represent discrete proximity regimes (McAlpine et al., 2015; Cai et al., 2025; Rolfe et al., 2026; Wallmo and Lew, 2015), or define jurisdictions and compare groups to assess how values differ across administratively or socially meaningful spatial partitions (Perry et al., 2014). In this study, we build on this broader literature by investigating both distance decay and spatial clustering in preferences for Oregon's MRS, thereby aligning with the main strands of existing methods.

Also, a wide range of ecosystem services and environmental goods have been examined through the lens of spatial preference heterogeneity, including forest management and conservation programs (Czajkosvky et al., 2017; Eusse-Villa et al., 2025; Budzinsky et al., 2018), water-quality improvements (Shaafma et al., 2013; Bateman et al., 2006; Brouwer et al, 2010; Olsen et al., 2020), endangered species recovery and biodiversity protection (Johnston and Ramacharadan, 2014; Johnston et al., 2015; Johnston et al., 2017; Wallmo and Lew, 2015; Yao et al., 2019), and other public environmental goods (Ferrini and Fezzi et al., 2012; Faccioli et al., 2020). The empirical literature also suggests that the spatial patterning of values can differ across ecosystem-service types, particularly when contrasting use-related benefits with non-use motives and local/global benefits (Bateman et al., 2006; Wallmo and Kosaka, 2017; Shaafma et al., 2003; Johnston et al., 2015; Olsen et al., 2020; Hein et al., 2016; Fisher et al., 2009; Dallimer et al., 2014; Torres et al., 2015; Bakhtiari et al., 2018). Marine and coastal studies similarly report heterogeneous

and segmented support for protection measures, often indicating that individuals who live closer to the coast - or who are otherwise more connected to coastal environments - value marine protection differently (Sagebiel et al., 2016; Johnston & Ramachandran, 2014; McAlpine et al., 2025). This evidence implies that preferences for marine reserve policies are plausibly spatially patterned, but it remains unclear whether these preferences form distance decay, smooth spatial gradients, discrete clusters, or more complex spatial structures across the broader population.

In this paper, we implement a choice experiment to evaluate preferences for changes in MRS policy in Oregon, focusing on MR size, blue carbon capture, and coastal jobs. These attributes differ in their underlying economic nature and policy characteristics, including the extent to which benefits generate use versus non-use values, local or global benefits, and the degree of substitutability with nearby alternatives.

We evaluate preference heterogeneity using a latent class model (LCM), incorporating travel time (a distance/proximity measure) and attitudinal factors as covariates to examine how these drivers shape distinct categories of MRS preferences. In addition, we apply two-dimensional spatial generalized additive models (GAMs) and Global Moran's I statistics to assess whether preferences exhibit spatial autocorrelation and to characterize how preference patterns are distributed across space. MRS size and job changes tends to generate more locally salient benefits, whereas blue carbon mitigation a more diffuse/global benefit. This motivates evaluating preferences and acceptability across Oregon residents in two ways: i) by estimates of probability of belonging to preference segments, therefore allowing to analyze if environmental attitudes and worldviews that shape valuation cluster geographically, and ii) by marginal utility of individual attributes, allowing that each element of the policy bundle exhibit distinct, and empirically testable, spatial and preference patterns. Building on work documenting distance-related patterns in Oregonians' preferences for Marine Reserves (Perry et al., 2014; McAlpine et al., 2025), we extend this literature by going beyond proximity to the coast and testing for statistically significant spatial clustering in preferences. Methodologically, we further contribute by, to our knowledge, being the first paper to combine LCM with spatial GAMs estimation/mapping and Global Moran's I tests, using these complementary tools to corroborate whether spatially structured patterns in predicted preferences provide additional inferential evidence beyond mean estimates from LCM alone.

2 Literature Review

To situate our hypotheses, we review stated-preference studies that document how environmental preferences vary across space and across people. In particular, we organize the evidence around three mechanisms that motivate our empirical strategy: (i) proximity and opportunity sets (distance decay), (ii) spatial dependence beyond dis-

tance (clustering/autocorrelation), and (iii) how the spatial incidence of benefits (local vs diffuse/global) conditions the expected spatial structure of preferences. Together, this literature clarifies when spatially explicit modeling improves welfare inference and when observed spatial patterns may reflect both economic mechanisms and geographically patterned beliefs.

First, a large stated-preference literature shows that welfare measures for site-based environmental changes often exhibit distance decay, particularly when benefits are tied to access, use, or locally experienced improvements. Bateman et al. (2006) highlights that distance decay is expected for resource with use values. Instead of aggregating values using political jurisdictions and reporting means, they found that incorporating proximity often matters for non-market values when estimating policy benefits. Johnston et al. (2018) embed distance directly in a meta-regression using geospatially constructed measures and show that distance decay is associated with systematic changes in mean WTP, as distance from the affected resource increases. In a study valuing changes in bird populations and landscapes (local benefits) using a survey with individualized choice maps, Badura et al. (2020) report a statistically significant log(distance) effect (for the subsample shown maps with distances) of about 4.23, indicating distance decay. Schaafsma and Brower (2020) found that distance decay occurs with water bodies and is mediated by substitutes and by users of the environmental good in question. They build the spatial structure directly into the discrete-choice model (mixed logit). Ruiz-Frau (2019) used an LCM to evaluate Marine Reserves, supporting the idea that societal segments differ in their support and stating that proximity/connection to the coast is part of what differentiates segments. Not by distance decay, but by defining a geographical stratum, McAlpine et al. (2025) document systematic geographic differences in support for Marine Reserves in Oregon, where the most favorable are people living closest to a reserve and non-coastal residents in Oregon's most populated inland region. The least favorable are coastal residents who live on the coast but farther from a reserve. Perry et al. (2014), in a social survey study, operationalize proximity to Oregon's marine reserves using "communities of place within 10 miles" versus the "rest of the coast," generating comparative results by group rather than presenting a welfare model. They find that self-assessed knowledge and factual knowledge did not differ between the proximate group ($\leq$ 10 miles) and more distant coastal residents.

Beyond distance, preferences may exhibit spatial dependence, implying clustering rather than independent noise. Johnston and Ramacharadan (2014), in a two-step spatial approach, analyzed geocoded stated preferences for river restoration and argue that assuming spatial homogeneity and continuous distance decay can miss significant non-conscious patches in estimating preferences. Therefore, he motivates the use of spatial statistics approaches to detect clustering beyond distance to environmental goods when evaluating environmental preferences. Czajkowski et al. (2017) combined LCM, spa-

tial lag modeling, and regression-kriging interpolation and found distance decay, spatial autocorrelation, and geographic clustering in WTP and class probabilities for forest-management attributes in Poland (protection, litter, and tourism infrastructure). Based on these results, they argue for location-targeted policy communication and emphasize that substitution across sites can shape preferences when the environmental good is widespread rather than site-specific. Campbell et al. (2009), using the same methods but deriving estimates from a mixed logit model, identified geographical clusters in preference segments for rural landscape improvements. For soil ecosystem services in an Italian region, analyzed by Eusse-Villa et al. (2021), preferences do not exhibit significant global spatial autocorrelation, implying heterogeneity without global clustering.

The spatial incidence of benefits from environmental goods can also condition the strength and form of the spatial preference structure. Dallimer et al. (2015) confirmed the existence of a statistically significant "own-country" premium, in which people are willing to pay more when the same ecosystem service improvement occurs in their country of residence. Also, premiums are higher for local services, such as habitat conservation and landscape preservation, than for global services (such as carbon capture). Granado-Diaz et al. (2019) estimated that WTP for biodiversity and soil sequestration increases when respondents are "near" larger olive-grove areas, whereas WTP for carbon sequestration shows the opposite pattern. In Torres et al. (2015), people are willing to pay more for forest offset projects closer to home, and they link this to local co-benefits and local identification/visits/knowledge. Bakhtiari (2018) provides evidence that diffuse/global framing does not imply "no spatial structure". Even for a good they explicitly discuss as a global public good (biodiversity), they show statistically significant distance and jurisdiction/country effects. Faccioli et al. (2020), in an embedded space Discrete Choice Experiment (DCE) using a hybrid mixed logit model, found that geographically clustered attitudes and place identity can also shift the valuation of peatland restoration, supporting the claim that non-distance drivers can underlie spatial variation. They argue that even diffuse benefits (e.g., carbon storage and biodiversity) can exhibit spatial patterning because identity and attitudes are themselves spatially structured—an effect not captured by distance measures alone. Hoyos et al. (2015) and Eusse-Villa (2021) provide evidence that attitudes and beliefs can be spatially structured and shape the valuation of ecosystem services.

Taken together, the literature suggests that proximity often matters for site-based improvements, leading to distance decay that is especially relevant when benefits are experienced locally or mediated by access and substitutes. Also, spatial autocorrelation on welfare measures is often present and policy-relevant, as values and even segment membership can cluster geographically. However, such spatial structure is not guaranteed: preferences may be spatially random, distance effects may be weak once covariates are controlled for, and observed clustering can vary by attribute depending on whether

benefits are local (e.g., access or use values) versus diffuse (e.g., global climate benefits), as well as whether attitudinal worldviews are themselves geographically patterned. This motivates the use of flexible spatial diagnostics, such as GAM surfaces and global autocorrelation tests, as a complement to discrete-choice segmentation. The literature suggests that the expected spatial structure of attribute valuations can differ by attribute type, though with caveats. Locally salient attributes tend to generate stronger distance decay, whereas diffuse/global services may exhibit weaker distance decay on average, and any remaining spatial autocorrelation may reflect jurisdictional factors and spatially patterned attitudes and place identity.

3 Data

We begin this section with a discussion of the survey instrument and the embedded choice experiment designed to elicit household preferences for different Oregon's MRS scenarios. We then discuss the data and associated summary statistics.

The survey instrument developed for this study uses a standard approach to understand Oregonian's preferences for changes in the nearshore spatial management (c.f., Nguyen et al. (2023), Riegel et al. (2023), and Chen et al. (2023) for examples of similar survey approaches). The survey design follows guidelines developed by Johnston et al. (2017)[1]. To ensure the capture of key dimensions and the elicitation of a wide range of values associated with Oregon's nearshore spatial management, we carefully designed and operationalized the questions, and incorporated feedback from experts and stakeholders through three focus groups[2]. We then conducted a pilot evaluation of the survey instrument, created in Qualtrics XM, in June and July 2023, first informally with graduate students and professors at Utah State University, and later with a subset of 100 survey panel participants residing in Oregon and recruited by Qualtrics. Survey content was revised based upon these pilots. The final version of the survey was administered between August and December of 2023 to another 373 survey panel participants and an additional 68 respondents recruited via mailed invitations from a sample frame provided by Dynata, Inc. following the Tailored Design Method (Dillman et al., 2014)[3].

The survey starts with an informed consent letter followed by questions regarding the

[1]Our survey instrument is available upon request from the corresponding author.

[2]Feedback from these groups on a draft of the survey helped guide the research team in clarifying the survey's introductory materials. For example, focus group participants were generally unfamiliar with the terminology "carbon sequestration", which was therefore replaced with "capture and store carbon" in the survey's final version.

[3]The first round of testing incorporated cognitive interviews with potential participants, as recommended by Johnston et al. (2017). We altered the cost interval and the blue carbon levels after the 100 piloted surveys, and made no changes to the survey design thereafter. As a result, the 100 piloted surveys were discarded. Our survey pretesting procedure follows the approaches used in Wallmo and Edwards (2008), Chen et al. (2023), Meles et al. (2023), Aanesen et al. (2023), and O'Dell et al. (2025).

participant's use of and familiarity with Oregon's coast. Next, we provide background information on the reserves, which included the the map displayed Oregon MRS as a point of reference, as well as an enumeration of the benefits and costs associated with preserving, monitoring, and administering the reserves, including the enforcement of regulations[4].

Because it addresses the issue of blue carbon within a coastal ecosystem framework – an issue that is likely novel to many of our participants – the survey provided the following information to participants,

> *"Currently, human activities release more carbon into the atmosphere than they capture and store. Reducing the risks of extreme weather, sea level rise and wildfires requires capturing and storing carbon dioxide. Marine and coastal ecosystems including kelp forests, seagrass beds, and tidal marshes can remove carbon from the atmosphere. These ecosystems can capture and store more carbon per area than most types of land-based forests. Restoring and maintaining these ecosystems could help reduce carbon dioxide in the atmosphere."*

To gauge survey participant views on climate change, we utilized a modified climate urgency scale based on Nawaz et al. (2023). Two attention checks were utilized for quality control. Following Oh and Gramig (2023), participants who answered either of these two checks incorrectly were excluded from analysis[5].

The choice experiment was presented next, followed by a series of statements gauging the participants' (1) environmental worldviews, including statements like, "Nature is important mostly because of what it can contribute to human well-being," (choose from "strongly disagree" to "strongly agree"), (2) social values, with statements such as, "Freedom – liberty for everyone to think and act as they consider appropriate," (choose from "not important" to "extremely important"), and (3) individual values, with statements such as, "Tradition is important to [me]. [I] try to follow the customs handed down by [my] religion or [my] family," (choose from "not like me at all" to "very much like me").[6]

[4]Benefits are listed as: (1) protection and restoration of habitat and species, (2) scientific research, educational, and tourism opportunities, and (3) potential fisheries enhancement, such as by protecting fish nursery grounds. Costs are listed as: (1) reduced access for commercial and recreational fishing, with possible displacement to other areas, (2) financial costs of administering the nearshore spatial management, including monitoring human activities and enforcement of regulations, e.g., only specific types of fishing gear are allowed in certain MPAs, and (3) financial costs of ecological monitoring, e.g., scientists track how habitat and fish populations change over time in the nearshore spatial management.

[5]Scarpa et al. (2009) solely removed respondents who chose only the status quo. Riegel et al. (2023) included three "red herring" questions to test whether respondents read the provided information text; respondents were screened out if they replied incorrectly to two out of the three questions. Respondents were also screened out due to speeding concerns, as well as via failing an attention test similar to ours. This is in keeping with Malone and Malone and Lusk (2018) "trap questions," which led to the conclusion that "participant inattention can substantially bias parameter estimates and subsequent policy recommendations (page 219)".

[6]Each question also included "not sure" as a choice option. Individuals who answered this alternative were considered as "NAs" and excluded if the question was included as a variable of analysis.

Standard socio-demographic questions comprised the final section of the survey.

We developed an MNL d-efficient fractional factorial design for the choice experiment (which corresponds to a 3x3x4x7 full factorial design matrix, refer to Table 1 below) using the Ngene software package, with pilot design priors based upon our previous study and final design priors based upon pilot results[7]. The design includes 16 choice tasks divided into four blocks, with each respondent completing four choice tasks in randomly-assigned blocks. The design's d-efficiency score after eight iterations on the final experimental design was 1.373.

Each choice situation consists of two separate policy-change alternatives (alternatives 1 and 2) and an "opt-out" status quo option. Each of the alternatives is comprised of four separate attributes with Table 1 providing their levels and variable names for ensuing empirical analysis.

[INSERT TABLE 1 HERE]

The attribute *MRS* corresponds to percent changes in MRS size (ranging from -50% to 50%), *Carbon* measures the percentage change in storing and capturing carbon from the marine ecosystems (ranging from zero to 100%); *Jobs* refer to changes in the number of jobs related to fishing, blue carbon and tourism along the Oregon coast. Local and state government subsidies for attracting and retaining businesses[8] reflect public support for job creation in their communities. The present focus is not on the social welfare effect of incentive programs, but on public values for job creation. Research indicates positive resident WTP for jobs, including in coastal and marine contexts (Xuan et al. 2021).

Carbon reflects the percentage change in carbon sequestration relative to the status quo. This metric is consistent with other studies (Bocci et al. 2020; Brodersen et al. 2024) and a reflection of the complexity and uncertainty in calculating carbon sequestration (Börger et al. 2014; Luisetti et al. 2014; Macreadie et al. 2019). The percentage change metric also reflects the project goal of providing an initial assessment of public preferences, rather than an estimate of WTP per ton of carbon dioxide equivalent for use in benefit-cost analysis.

Lastly, *Cost* denotes the added annual household cost in the form of increased state income taxes, for the next 10 years. This type of non-voluntary payment mechanism is known to enhance a DCE's incentive compatibility and payment consequentiality (Carson and Groves, 2007; Johnston et al., 2017)[9].

The marine reserve system and blue carbon attributes both reflect potential changes in nearshore management. However, they are independent of each other in presentation,

[7]For information on Ngene see https://www.choice-metrics.com.

[8]See, for example, `https://www.mckinsey.com/industries/public-sector/our-insights/how-state-and-local-governments-win-at-attracting-companies/`

[9]Recently, Cunha-e-Sá et al. (2025) find that respondents were more willing to contribute voluntarily to a fund protecting mainland Portugal's vulnerable coastal region when a crowdfunding campaign was coupled with a provision-point mechanism and a money-back guarantee.

in practice, and statistically. Visually, the two attributes are presented separately, both in the background information (Figure 1: Panel A) and in each choice task (Figure : Panel B). The background information makes clear that blue carbon would occur across Oregon's territorial waters, with no link to the marine reserve system. In practical terms, monitoring occurs in marine reserves, but no active management is allowed. Therefore, blue carbon increases due to policy and management changes would occur outside marine reserves.

Statistically, the MRS and Carbon attribute levels took the same value (i.e., both "no change" or both "50% increase") in only 12% of the choice alternatives seen by respondents. In 38% of the alternatives the MR and Carbon attribute levels had inconsistent signs (i.e., "50% decrease" for MR with "no change" or an increase in Carbon), with the remaining alternatives having the same sign but differences in magnitude across levels for the two attributes. The highly significant attribute coefficients (results below) indicate that respondents accurately understood the attribute levels and thus the independence of these two attributes.

An example choice situation is presented in Figure 1 (Panel B), prefaced by an accompanying information blurb that lends context to the ensuing choice situations (Panel A).

[INSERT FIGURE 1 - PANEL A/PANEL B HERE]

In this example, Option 1 is associated with a 50% decrease in the size of Oregon's marine reserves, 200 new jobs gained, no change in the system's blue carbon capacity, and an associated increase in annual household cost of $400. In contrast, Option 2 is associated with a 50% increase in the size of Oregon's marine reserves, no change in jobs, a 100% increase in the system's blue carbon capacity, and an associated increase in annual household cost of $100. The respondent then selects either the current situation (no change), Option 1, or Option 2, and indicates on a sliding scale from zero to 100% the certainty that the response reflects his or her preferences across the three alternatives.

Our DCE was designed to control for a host of factors affecting the accuracy, or incentive compatibility, of participant responses. These include (1) controls for both policy and payment consequentiality, (2) controls for potential biases associated with the survey's hypothetical construct, (3) availability of information necessary for respondents to understand exactly what they are being asked to value (e.g., Panel A of Figure 1), and (4) controls for both survey and attribute non-attentiveness. The survey also includes controls for response uncertainty (e.g., bottom of Panel B of Figure 1), which enables us to perform a re-coding of the data to obtain WTP estimates in the context of respondent uncertainty (c.f., Lundhede et al., Li and Ando, 2023). To address policy consequentiality, we included statements and cheap talk to address potential hypothetical bias.

Respondents were excluded if they (1) protested the choice task, or did not respond

to a protest survey item[10], or (2) completed fewer than half of the full list of indicators for one or more of the included attitudinal characteristics – individual, social and environmental values. Excluding "protest voters" is mainly about measurement validity intending to reduce the influence of protest or payment-vehicle rejection responses — i.e., cases where the observed choices may primarily reflect objection to the financing mechanism rather than tradeoffs over the policy attributes in the choice experiment[11]. After these exclusions, the analyzed data were from 371 respondents, combined across the two sets, and a total of 1,484 choices. The data were weighted by gender to reflect population values in Oregon.

As indicated in Table 2, the US census estimates for several of our sociodemographic variables fall within the 95% confidence intervals associated with their corresponding sample means. For example, the census estimate for middle-aged Oregonians (*midage*) of 40.6% falls within the 95% confidence interval for our sample mean of 40.4%. The census estimates for percentage of respondents who are Black, Asian, Pacific, or Other similarly fall within their respective 95% confidence intervals, as do the estimates for respondents with some college (*college*) or a graduate degree (*graddeg*), and those reporting high-middle incomes (*himidinc*). It is also interesting to note that the census estimate for employed individuals of 60% falls within the aggregate confidence interval for full- and part-time respondents (*workft* and *workpt*, respectively).

[INSERT TABLE 2 HERE]

Further, as Table 2 indicates, while census estimates for *senior* and *bachdeg* come close to falling within their associated confidence intervals, those for *gender, white, latino, highinc, midinc*, and *urban* do not. In particular, our sample includes higher percentages of males, whites, and middle-income earners than the general Oregon population, and lower percentages of Latinos, high-income earners, and urban households. Our empirical results should therefore be interpreted with these features of the data in mind. The data used for analysis were weighted by gender to better reflect its population distribution. Summary statistics for each of the climate urgency, relational, and recreational variables and percentage comparisons are included in Appendix 5.

Each respondent reported a five-digit ZIP code, which we linked to Oregon ZIP-code polygons in ArcGIS Pro. Using Network Analyst's OD Cost Matrix, we computed the minimum driving time from the centroid of every ZIP polygon to the nearest point on the coastline. We used the travel mode: Driving Time and time setting: Not using time of day, i.e., no traffic. All layers were projected to NAD83 / UTM Zone 10N (EPSG:26910).

[10] After completing the four choice tasks, respondents indicated on a scale from 1 = strongly disagree to 7 = strongly agree the extent to which they agreed with the statement "I do not believe the government should fund marine reserves with our taxes." Respondents with missing values or with responses of 6 or 7 to this statement were excluded from the analysis.

[11] Resulting welfare estimates should be interpreted as conditional on willingness to consider tax-based funding.

Total Travel Time output in minutes was joined back to the survey microdata by ZIP.

4 Methodology

4.1 Factor Analysis and Factor Scores

We use factor analysis to summarize respondents' attitudinal/worldview measures into a small set of latent constructs. The survey contains 33 Likert-scale items (7 response categories), which we first group conceptually into blocks reflecting hypothesized constructs and then refine empirically. We treat these items as ordinal and summarize their dependence structure using a polychoric correlation matrix, which is appropriate when each observed ordinal response is viewed as a categorized version of an underlying continuous latent response [12].

Let x_{ij} denote respondent $n = 1, \ldots, N's$ observed response to item $j = 1, \ldots, P$ (here P=33). We use principal compenent analysis (PCA) and parallel analysis to determine the number of common factors Q. Parallel analysis compares eigenvalues from the observed data to eigenvalues from randomly generated data with the same N and P, retaining factors whose observed eigenvalues exceed the random benchmark (Hayton et al., 2004).

We then conducted an exploratory factor analysis (EFA) which is based on the common-factor model with Q factors (Jöreskog, 1970):

$$x_{nj} \;=\; \tau_j \;+\; \sum_{q=1}^{Q} \lambda_{jq} f_{nq} \;+\; \varepsilon_{nj}, \tag{1}$$

where τ_j is an item intercept (often omitted after centering/standardization), λ_{jq} is the loading of item j on factor q, f_{nq} is respondent n's score on factor q, and ε_{nj} is the unique (item-specific) component. The Uniqueness $\psi_j = Var(\varepsilon_{nj})$ corresponds to the variance of the unique component, being the share of item $j's$ variance not explained by the common factors.

Exploratory factor analysis EFA is estimated by fitting the model-implied covariance structure to the observed covariance (here, the polychoric correlation matrix). In matrix form,

$$\Sigma = \mathrm{Var}(x_n) = \Lambda\Phi\Lambda' + \Psi \tag{2}$$

where $\Lambda = [\lambda_{jq}]$ is the $P \times Q$ loading matrix, $\Phi = \mathrm{Var}(f_n)$ is the factor covariance

[12] Missing Likert-item responses were handled with Multiple Imputation by Chained Equations (MICE) using an ordinal proportional-odds model generating one completed dataset, iterated 10 times. MICE impute each variable conditionally on all others, cycling through these conditional models iteratively (fully conditional specification), repeating this cycle until the imputations stabilize. The completed data were then used to com-pute the polychoric correlations and estimate the factor model (Graham 2009; Enders, 2010; Van Buuren, 2018).

matrix, and $\Psi = \text{Var}(\varepsilon_n)$ is diagonal. Because the attitudinal dimensions are conceptually related, we allow correlated factors (i.e., Φ is not constrained to be diagonal) and apply an oblique rotation. Specifically, we use promax rotation, which yields a simple structure while permitting correlated factors (Horn, 1965).

Factor scores are not uniquely identified by Σ alone; therefore, after estimating $\hat{\Lambda}$, $\hat{\Phi}$, and $\hat{\Psi}$, we compute respondent-level factor score estimates $\hat{f}_{nq}$ using a scoring rule. We use Bartlett scores, which weight items by their estimated uniqueness, downweighting noisier items and yielding scores with high validity (Grice, 2001; DiStefano et al., 2009).

We refine the scale using a combination of EFA diagnostics - Tucker–Lewis Index (TLI), Cronbach's alpha, and the mean square residua;l (RMSR) - and confirmatory factor analysis (CFA), applied to the full sample and to train/test splits as a robustness check (Fabrigar et al., 1999).[13] Because the attitudinal dimensions are conceptually related, we allowed the factors to be correlated and applied an oblique rotation. Specifically, we used promax rotation, which yields a simple structure while permitting correlated factors (Horn, 1965).[13].

4.2 Latent class model (LCM)

Our segmentation analysis is grounded in random utility theory (Lancaster, 1966; McFadden, 1974), where respondent n chooses the alternative j in the DCE that yields the highest utility in choice task t. Formally, conditional on latent class c, utility from alternative j is (Boxall and Adamowicz, 2002; Hensher et al; 2015);

$$U_{njt|c} \;=\; \beta_c' x_{njt} \;+\; \varepsilon_{njt}, n = 1, ..., N; j = 1, ..., J; t = 1, ..., T, \tag{3}$$

where x_{njt} is a vector of alternative attributes related to: specific level of the status quo indicator (which is a constant), the percentage change in MRS size and in blue carbon, unit changes in ocean and cost. β_c is the class-specific preference parameter vector, and ε_{njt} is an i.i.d. extreme-value (type I) error term.

Under this assumption, the probability that respondent n chooses alternative j in task t, conditional on class c, takes the multinomial logit form:

$$L_{njt|c} \;=\; \frac{exp\left(\beta_c' x_{njt}\right)}{\sum_{i=1}^{J} exp\left(\beta_c' x_{nit}\right)}, n = 1, ...N, j = 1, ...J, t = 1, ...T, and\ c = 1, ...C. \tag{4}$$

[13]Factor analysis was estimated on all respondents who provided usable attitudinal data (N = 441) to maximize stability of loadings and factor correlations, consistent with recommendations for EFA with Likert indicators (Fabrigar et al. 1999; MacCallum et al. 1999). For the latent class model, we imposed a stricter rule and re-tained only respondents who (i) did not state a protest belief and (ii) answered at least half of the indicators in each battery (N = 371), to ensure reliable individual factor scores as covariates. We verified that the factor solution from N = 441 and the solution from the analysis sample (N = 371) were factorially equivalent (Tuck-er's $\phi \geq 0.99$; Lorenzo-Seva and Ten Berge, 2006), so using the 441-based factors in the LCM does not change substantive conclusions.

Assuming independence across tasks conditional on class, the probability of respondent n's observed sequence of choices, conditional on class c, is

$$P_{n|c} \;=\; \prod_{t=1}^{T} L_{ny_{nt}t|c}, n = 1, ...N, t = 1, ...T, and\ c = 1, ...C, \tag{5}$$

where y_{nt} denotes the chosen alternative in task t, i.e: $y_{nt} \in 1, \ldots, J, ynt = j$ iff respondent n chooses alternative j in task t.

In the LCM, respondents are probabilistically sorted into one of the C pre-defined latent classes. We parameterize the prior probability that respondent n belongs to class c using a multinomial logit "class membership" function:

$$H_{nc} \;=\; \frac{exp\left(\gamma_c' z_n\right)}{\sum_{r=1}^{C} exp\left(\gamma_r' z_n\right)}, n = 1, ...N, c = 1, ...C, \tag{6}$$

where z_n is a vector of respondent covariates and γ_c is the corresponding class-specific coefficient vector (Greene and Hensher, 2003; Hensher et al., 2015). Here, z_n includes the extracted attitudinal/factor score measures and travel time to the coast, which enter the class membership function to explain systematic designation into preference segments [14].

Because we do not observe which class a respondent belongs to, we compute the likelihood by summing over classes, weighting each class's choice probability by the probability of belonging to that class. The unconditional probability of respondent $n's$ observed choice sequence:

$$P_n \;=\; \sum_{c=1}^{C} H_{nc} P_{n|c} = \sum_{c=1}^{C} \left[\quad \frac{exp\left(\gamma_c' z_n\right)}{\sum_{r=1}^{C} exp\left(\gamma_r' z_n\right)} \quad \right] \prod_{t=1}^{T} \frac{exp\left(\beta_c' x_{njt}\right)}{\sum_{i=1}^{J} exp\left(\beta_c' x_{nit}\right)} \tag{7}$$

Model parameters are estimated by MLE maximizing $\sum_{n=1}^{N} \ln(P_n)$. The LCM was estimated in Latent Gold 6.1[15].

4.3 Global Moran's Test for Spatial Autocorrelation

To assess whether estimated preference measures exhibit global spatial correlation and clustering, we computed the Global Moran's I statistic for (i) posterior latent-class membership probabilities and (ii) individual-level attribute coefficients (posterior weighted expectations) using a k-nearest neighbors (k-NN) spatial weights matrix with k=8 built on ZIP-code centroids. Coordinates were projected to a metric CRS and slightly jittered (10 m) to separate coincident points, and the neighbor graph was forced to be connected.

[14] Covariates are effects coded.

[15] The LCM framework above nests the equality-constrained latent class (ECLC) approach I used previously in Essay 1 and 2. In ECLC, some utility parameters are set to zero within particular classes to represent inferred non-attendance, while the remaining (attended) parameters are constrained to be equal across classes. In Chapter 3, by contrast, we use an unconstrained LCM for segmentation: preference parameters β_c are freely estimated by class, while observed covariates (factor scores and travel time) shift class membership probabilities through H_{nc}.

We report both the analytical (normal approximation) test and a Monte Carlo permutation test with 999 random reallocations.

To perform Global Moran's I statistic we start by defining the weight matrix (Ciff and Ord, 1973; Plant, 2018). Let $n = 1, \ldots, n$ an index of observations located at ZIP-centroid coordinates, and let j an index of potential neighbors of i. For a given variable of interest x_i (e.g., the posterior probability that respondent i belongs to latent class c, or an individual-level coefficient for an attribute), we define a k-nearest-neighbors set $N(i)$ with k=8 and symmetrize it so that neighborhood relations are mutual (if i selects j, then j also selects i). First, we construct a binary adjacency matrix $W = [w_{ij}]$, where

$$w_{ij} = \begin{cases} 1 & \text{if } j \in N(i), \\ 0 & \text{otherwise,} \end{cases} \quad \textit{and } w_{ii} = 0. \tag{8}$$

We then define the row-standardized spatial weights matrix $W^{(r)} = \left[w_{ij}^{(r)}\right]$ [16] by dividing each neighbor's weight by the sum of all neighbor weights in that row and normalizing each row:

$$w_{ij}^{(r)} = \frac{w_{ij}}{\sum_{j=1}^{n} w_{ij}} = \begin{cases} \frac{1}{|N(i)|} & \text{if } j \in N(i), \\ 0 & \text{otherwise.} \end{cases} \tag{9}$$

By construction, as row standardization implies $\sum_{j=1}^{n} w_{ij}^{(r)} = 1$ for each i. Let

$$S_0 = \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}^{(r)} \tag{10}$$

denote the sum of all weights under row-standardization and in the absence of isolates (no observations with zero neighbors), therefore $S_0 = n$.

The Global Moran's I statistic measures whether deviations from the global mean of a scalar outcome co-vary across neighboring locations (Moran, 1950; Cliff and Ord, 1981; Anselin, 1988). Let x_i denote the outcome of interest observed at location i (e.g., a posterior class-membership probability or an individual-level attribute coefficient). Define the sample mean $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and deviations $z_i = x_i - \bar{x}$. Then, the Global Moran's I statistic is:

$$I \;=\; \frac{n}{S_0} \frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}^{(r)} z_i z_j}{\sum_{i=1}^{n} z_i^2} \tag{11}$$

Values $I > 0$ indicate clustering of similar values (high–high and low–low), $I < 0$ indicates spatial dispersion (checkerboarding), and $I \approx 0$ is consistent with spatial randomness (Moran, 1950; Cliff and Ord, 1981). For inference, we report analytical p-values, assuming Moran's I (and/or its standardized form) as asymptotically normal as n

[16] (r) denotes row-standardized weights.

grows. Under the null hypothesis of spatial randomness, the expectation is $E[I] = -\frac{1}{n-1}$, and the variance $Var(I) = E\left[(I - E[I])^2\right]$ is computed as a function of $W^{(r)}$ and the sample moments of x (Cliff and Ord, 1981; Anselin, 1988).

The standardized test statistic is:

$$z = \frac{I - E[I]}{\sqrt{Var(I)}}. \tag{12}$$

Because the exact finite-sample distribution of I can deviate from the normal approximation and to improve robustness, we also compute a permutation-based p-value using 999 random reallocations of the observed values across locations while holding $W^{(r)}$ fixed (Anselin, 1988; Plant, 2019). Let B=999. For each permutation $b = 1, \ldots,$B, we randomly permute x to obtain $x^{(b)}$, compute $I^{(b)}$ using the same formula above, and compare to I_{obs}. For a one-sided "greater" test, the Monte Carlo p-value is:

$$p_{\text{perm}} = \frac{1 + \sum_{b=1}^{B} 1(I^{(b)} \geq I_{\text{obs}})}{B + 1} \tag{13}$$

Global Moran's I statistic is implemented in R using spdep (Bivand et al., 2013).

4.4 Spatial Generalized Additive Models (Spatial GAMs)

We test for a smooth two-dimensional spatial surface in class-membership probabilities and estimated preference coefficients using GAMs (Wood 2003; Wood 2017). A GAM extends a a generalized linear model (GLM) by replacing linear effects with smooth functions of covariates. In general form, for observation $i = 1, \ldots, n$,

$$g(\mu_i) = \beta_0 + \sum_{q=1}^{Q} f_q(X_{qi}), \tag{14}$$

where $\mu_i = E[Y_i \mid X_i], g(\cdot)$ is a link function and each $f_q(\cdot)$ is a smooth function.

In our application, Y_i denotes either the posterior probability that individual i belongs to class c or the posterior mean of individual i's utility coefficient for attribute k, with observed value y_i. Each observation i is associated with a point location given by ZIP-centroid planar coordinates (u_i, v_i), obtained after projecting geometries to a metric CRS (EPSG:5070). For each outcome, we fit a bivariate spatial GAM of the form

$$g(\mu_i) = \beta_0 + s(u_i, v_i), \tag{15}$$

where s is a smooth surface function over space. Defining Gaussian errors and an identity link, $g(\mu_i) = \mu_i$, then the conditional mean is modeled directly on the original scale:

$$y_i = \beta_0 + s(u_i, v_i) + \varepsilon_i,\ \ \varepsilon_i \sim N(0, \sigma^2) \tag{16}$$

The bivariate smooth term $s(u_i, v_i)$ is represented using a thin plate regression spline basis (Wood, 2003) and k controls the basis dimension (maximum flexibility) of the smooth. We define the model with K=80 maximum basis dimensions.

Conceptually, the bivariate smooth can be written as a basis expansion

$$s = s(u_i, v_i) = \sum_{m=1}^{k} \theta_m b_m(u_i, v_i), \tag{17}$$

with coefficients θ_m estimated under a smoothness penalty (to prevent overfitting). Here b_m denotes the $m - th$ thin plate regression spline basis function which is a predetermined smooth function of (u, v) produced by the reduced-rank thin plate spline construction (a truncated eigenbasis of the thin plate penalty), with $K \ll n$. Thus, the coefficients θ_m are estimated from the data, while the basis functions b_m are generated algorithmically and are not written in closed form one-by-one. Thin plate splines are built from a smoothness idea: among all surfaces that fit the data, choose the one that minimizes a tradeoff between fit (small residuals), and wiggliness (a penalty on curvature/complexity of the surface). A thin plate spline is just a flexible "trend surface" over geography that is allowed to bend, but is penalized for bending too much. It's like fitting a smooth blanket over points in space. The actual smoothness is primarily controlled by the smoothing parameter λ via the penalty described next. Estimation solves a penalized least squares problem of the form:

$$\min_{\beta_0, \boldsymbol{\theta}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{m=1}^{k} \theta_m b_m(u_i, v_i) \right)^2 \ + \ \lambda J_{p,d}(g), \tag{18}$$

where $J(s)$ is a roughness measure (how "bendy" the surface is) and λ controls the smoothness. We define the code to estimate λ using a restricted maximum likelihood method (REML). Wood (2003) defines the thin-plate roughness functional (for dimension d and derivative order p) as

$$J_{p,d}(s) = \int \cdots \int \sum_{\substack{\alpha_1 + \cdots + \alpha_d = p \\ \alpha_j \geq 0}} \frac{p!}{\alpha_1! \cdots \alpha_d!} \left(\frac{\partial^p s}{\partial x_1^{\alpha_1} \cdots \partial x_d^{\alpha_d}} \right)^2 dx_1 \cdots dx_d. \tag{19}$$

which penalizes squared $p - th$ order derivatives (curvature/wiggliness of the surface). In our spatial case, d=2 (two coordinates), and $mgcv's$ thin-plate construction corresponds to penalizing second-order curvature (a specialization of the expression above to $d = 2, m = 2$) as

$$J_{2,2}(s) = \iint \left[\left(\frac{\partial^2 s}{\partial u^2} \right)^2 + 2 \left(\frac{\partial^2 s}{\partial u \, \partial v} \right)^2 + \left(\frac{\partial^2 s}{\partial v^2} \right)^2 \right] du \, dv. \tag{20}$$

Estimation yields $\hat{\beta}_0$ and $\hat{\theta}$ (and thus $\hat{s}$) from the penalized criterion in (18), with λ selected by REML. For inference on the spatial trend in (16), we report: (i) the estimated effective degrees of freedom (*edf*) for the smooth surface $s(u_i, v_i)$, interpreted as the *model degrees of freedom used by the smooth* (i.e., the number of parameters the smooth effectively "spends" after penalization), so larger *edf* indicates a less constrained and more flexible fitted surface; and the approximate F-test and associated p-value for the null hypothesis $H_0 : s(u, v) \equiv 0$ (equivalently, $\theta \equiv 0$ in (17)). (ii) As a complementary model-comparison check, we perform an incremental fit test by comparing the spatial GAM in (16) to an intercept-only model $y_i = \beta_0 + \varepsilon_i$ via ANOVA, reporting differences in model degrees of freedom, residual sums of squares (or deviance), the F statistic, and its p-value.

Finally, we visualize the estimated spatial trend surfaces by predicting $\hat{y}(x, y) = \hat{\beta}_0 + \hat{s}(x, y)$ on a dense grid over Oregon (in EPSG:5070), yielding smooth "trend surface" maps. We implement GAMs using the *mgcv* package in R.

5 Results

5.1 Factor Analysis

Initially, we group surveyed questions (scale items) into separate blocks reflecting shared underlying latent constructs. We then apply PCA and parallel analysis (Hayton et al., 2004) to identify the optimal number of factors. Starting with 33 scale items, parallel analysis suggests retaining 5 factors (Figure 2). Each scale item is a statement where respondents assign their accordance using a 7-likert scale (see Appendix 1 for their respective definitions). Subsequently, we conduct EFA and CFA to refine the factor structure and confirm the final number of scale items. These analyses were conducted both on the entire dataset and separately on train-test splits (Fabrigar et al., 1999). After these procedures, we reduced the scale items to 18 elements, retaining the 5-factor solution.

[INSERT FIGURE 2 HERE]

The final Factor Analysis with respective loadings, utilizing promax rotation, is summarized in the Table 3. We named the factors according to the scale items selected. The first factor, "Climate Concern", strong and positively correlates with beliefs emphasizing the urgent need to address climate change, the destabilizing impact of human activities on ecosystems, and climate change's potential to cause mass extinction events. It negatively correlates with the belief that climate change poses no environmental risk. The

second factor "Anthropocentrism", correlates positively with beliefs that environmental protection serves human interests, improves humanity's well-being, and that resource management should primarily benefit humans, with plants and animals existing mainly for human benefit. "Eco-centrism" positively associates with strong emotional connections to nature, perceiving plants and animals as family, caring for unseen natural places and recognizing individual personality in plants and animals. The "Conservative" factor relates to adherence to rules, avoidance of improper behavior and a strong emphasis on security. Lastly, the "Equalitarian" factor positively aligns with ideologies supporting equal opportunities, economic security and moral conduct. Expectedly, the factor structure reflects the same grouping of the survey scale items. Appendix 2 provides the polychoric correlation matrix among scale items, demonstrating higher correlations within the same factor loadings. The correlation between the factors exceeds the ± 0.32 threshold at least once for two factors (Appendix 3), justifying the selection of a correlated promax rotation over an orthogonal rotation (Tabachnick and Fidell, 2007). We opted to use the Bartlett method for factor score extraction because it generates unbiased factor scores with strong validity (Grice, 2001; DiSte-fano et al., 2009).

[INSERT TABLE 3 HERE]

5.2 Segmentation Analysis

After extracting the factor scores, we incorporate them, along with travel time from the coast (the spatial variable) into the class membership function within the Latent Class Model (LCM). This approach identifies distinct respondent profiles among Oregon citizens based on these characteristics, generating specific estimates for Marine Reserve System (MRS) attributes.

Latent-class specifications from one to ten classes were compared using AIC3, BIC, AIC and Log-likelihood (Table 4). The lower the information criterion the better the fit and the results show that distinct models minimize different fit measures. We follow Dias (2006) considering AIC3 the best criterion that minimizes over-fitting in LCM models in comparison to AIC and BIC[17]. Therefore, the criterion is minimized at five classes (AIC3 = 2575), providing our baseline model.

[INSERT TABLE 4 HERE]

Table 5 summarizes the marginal effects coefficients[18] for the MRS choice attributes: status quo (SQ), MRS size (MRS), jobs (Jobs), blue-carbon (Carbon), and cost (Cost) [19]. We report marginal effects to compare coefficient magnitudes consistently, since Classes 2 and 4 have insignificant cost coefficients and therefore yield insignificant WTP estimates. The estimated class proportions are well dispersed (Class 1: 33%, Class 2: 22%, Class 3:

[17]Log-Likelihood does not penalize the parameter increase, so it is just shown for routine purposes

[18]Utility coefficients are presented in the appendix 4.

[19]Table 1 shows the attribute meanings and levels.

21%, Class 4: 19%, Class 5: 6%), yielding both sizeable "mainstream" segments and a smaller tail of re-spondents with distinct preferences.

In Class 1, SQ is strongly disfavored (large negative coefficient), indicating openness to policy change. This segment favor policies that deliver employment and carbon benefits rather than larger reserves per se. The marginal effects reinforce the picture: a unit increase in SQ significantly reduces the predicted choice probability of selecting this option by 29%, and a unit increase in Carbon and Jobs significantly increase choice probability in a lower magnitude (0.2% and 0.05% respectively). Looking for the covariates (factors), being concerned about the climate risks and close from the coast significantly reduces the odds of belonging to Class 1. We name this class as "Jobs-and-Carbon pragmatists" because it has a higher likelihood to represent inland respondents who supports tangible outcomes and an indifference for change in MRS. Class 2 also dislike SQ while MRS and Carbon are positive and significant. Cost is not significant, indicating that it is not an element well considered among respondents. People who have high climate concerns are more eco-centric, less conservative, and closer to the coast tend to characterize this segment. This is the coastal, climate-salient group that supports larger reserves and co-benefits, we call it "Size-Carbon environmentalists". Class 3 significantly rejects SQ and supports all the other attributes. However, Cost is negative and significant, indicating that this group is price sensitive. Marginal effects show that MRS, Jobs and Carbon have the largest positive effects in the sample among the groups that reject the SQ. Factors show overall low impact in determining this class except for lower conservatism, which indicates a policy-forward group not tightly tied to geography. We call this group "Ambitious Expansionists". In class 4, SQ shows positively, which means that people in this group are averse to policy change. Although Jobs carry a modest positive effect, people dislike Carbon and are indifferent to MRS and Cost. Citizens with lower climate concern, more anthropocentric and less equalitarian are more likely to be in this group. Here, preferences are for no change or limited change unless job gains are salient with skepticism of carbon benefits. This group is ideological concise, so we name it "Status-quo-job idealists". Class 5 has the strongest SQ preferences in terms of keeping the situation as it is. Citizens support MRS and Jobs, discount Cost, while are indifferent to Carbon. People with lower climate concern than the average, more equalitarian, less anthropocentric, and farther from the coast characterize this segment. This minority is characterized as: "Strong status-quo keepers". In broader terms, across the first three classes - roughly three fourths of the sample - the status quo is dispreferred and at least two program attributes are positively valued. The remaining quarter of the sample places greater weight on avoiding change, yet even these respondents accept specific benefit-oriented features, particularly jobs.

[INSERT TABLE 5 HERE]

5.3 Spatial Class Probabilities Analysis

We extract individual posterior class probabilities from LCM and evaluate whether class-membership probabilities exhibited spatial autocorrelation using the global Moran's I statistic (Moran 1950; Cliff and Ord 1981; Anselin 1988), with a K-nearest neighbors spatial weights matrix (k = 8) built on the ZIP centroids. The graph was forced connected and coordinates were slightly jittered (10 m) to separate coincident points. We report both the analytical z-test, based on the expected value and variance of Moran's I under randomization, and a permutation test with 999 random reallocations of the probabilities to locations. Table 6 shows significant positive spatial autocorrelation for classes 1, 2 and 5, indicating spatial clustering of those class probabilities beyond random variation. No significant spatial autocorrelation was detected for classes 3 and 4. Hence, we can affirm that some preference segments (not all) exhibit geographic clustering, which means that for these segments people from closer locations have higher probabilities to belong to them.

[INSERT TABLE 6 HERE]

We modeled class probabilities with spatial Generalized Additive Models (GAMs) predicting bi-variate smooth of coordinates on a plane analogous to longitude and latitude $s(u, v)$ and renormalized, so the summation of class probabilities is equal to one at each location (Leathwick et al., 2006; Wood, 2008). GAM diagnostics tests whether adding a smooth spatial surface to a regression for class probability reduces deviance - i.e., explains systematic spatial variation in class probabilities, leaving less unexplained variation. The diagnostics checks if a continuous spatial gradient improves model fit beyond a flat surface (null), which means that the surface is not constant and probabilities drift smoothly over geography.

Map surfaces show that patterns are clear (Figure 3): Class 1 "Jobs-and-Carbon pragmatists" is dominant in the interior and east portions of Oregon, aligning with longer travel distances and outcome-focused preferences, consistent with broad support for moving away from the status quo when policies deliver jobs and carbon gains, with little intrinsic demand for larger reserves. Class 2 "Size-Carbon environmentalists" concentrates along the west coast and urbanized coastal corridors close to Portland, consistent with higher Climate Concern and shorter travel times. Class 5 tends to be localized in the southwest and central-interior regions, where the map confirms a strong SQ premium but responsiveness to explicit reserve-size and job gains. Classes 3 and 4 "Ambitious Expansionists" and "Status-quo-job idealists" are more diffuse and homogeneous distributed, corroborating the global Moran's results.

[INSERT FIGURE 3 HERE]

The dominant-class map (Figure 3) [20] confirms these gradients: coastal counties con-

[20] This map elects for each grid unit the class that has the highest probability accordingly to the GAM

centrate toward Classe 2; most interior counties are dominated by Class 1, with islands of dominance of Class 5 and limited dominant Class 4 presence in the southwest.

[INSERT FIGURE 4 HERE]

We report GAM results related to diagnostic of smooth significance (edf, F, p), and incremental improvement vs. null ($\Delta Deviance$ and F-test from ANOVA) in Table 7. It shows a significant spatial structure for classes 1 and 2, smooths are significant ($p \ll 0.001$) and the spatial term improves fit vs. null ($p_{inc} \ll 0.001$). In class 3, there is no detectable spatial structure and for class 4 there is weaker evidence ($p = 0.082$). Nevertheless, incremental test is significant (p_{inc} =0.034), being suggestive but small and corroborating by the dominance map where only a small local part is suggested to present spatial structure. Class 5 presents unconventional results: over-wiggly smooth (huge edf=36.5) and big deviance explained ($p_{inc} \ll 0.001$), but the approximate F for the smooth is not significant ($p = 0.124$). That can happen probably because the basis dimension is too high for the number of observations (only 6% of the sample). Therefore class 5 presents a complex surface where the overall smooth test is not significant and we treat it cautiously.

Overall, we can conclude that the probability of belonging to most of the preference segment varies smoothly over space and a continuous spatial gradient improves model fit beyond a flat surface (null). For classes 1 and 2 we found strong evidence of $s(u, v)$ with edf>1, small p, and clear incremental fit vs the intercept-only model (ANOVA F-tests).

5.4 Spatial Attribute Analysis

[INSERT TABLE 7 HERE]

We've also conducted a spatial analysis for the individual attributes' utility coefficients from LCM which are the posterior expectations of β given the data, calculated by probability-weighted average of the class-specific vectors. Table 8 shows Global Moran's I results (also using k-NN, with k = 8). We find positive and significant spatial autocorrelation in Job and Carbon. SQ has a modest/borderline significance and MRS and Cost shows no evidence at 10% level.

[INSERT TABLE 8 HERE]

Table 9 shows the model-based spatial structure and diagnostics for spatial GAM. The smooth $s(u, v)$ is highly significant for MR ($p = 0.0048$) and Job ($p = 0.0005$), with the largest effect sizes regarding deviance explained (0.203 and 0.222) and R^2_{adj} (0.149 and 0.168). Carbon is borderline at 5% level on the smooth ($p = 0.053$) but still yields a significant incremental fit vs. null ($F_{inc} = 2.114$; $p_{inc} = 0.009$). SQ shows modest deviance explained and a nonsignificant smooth ($p = 0.178$), yet the null vs. spatial comparison is significant at 5% ($F_{inc} = 1.782$; $p_{inc} = 0.030$), indicating a small but detectable spatial

surface estimated.

component. Cost presents no-spatial smooth structure by both the smooth test and the incremental diagnostics ($p = 0.315$ and 0.104).

The disagreement in MRS being significant in GAMs but not in Global Moran's I might be expected because the tests answer different questions. Because they target different spatial findings, some divergence is expected. While Moran's I tests global autocorrelation around a constant mean, GAMs ask whether a smooth trend $s(u, v)$ explains variation and whether adding it improves fit. Also, non-monotone trends with local characteristics and with adjacent positive/negative lobes can cancel in Moran's I but are captured by a smooth surface. Whether Moran's result depends on the k-NN graph (k = 8) GAM adapts its effective scale via smoothing.

[INSERT TABLE 9 HERE]

In figure 5, the surfaces display coherent spatial gradients rather than noise, where each attribute is most/least valued on your GAM maps, where the color red indicates higher coefficient (higher marginal utility for that attribute) and white indicates lower/negative utility in the spectrum. The coefficients were standardized to admit a coherent variation to compare values among attributes.

Spatial dependence in the estimated coefficients is strongest and most robust for Jobs and Carbon; it is present but weaker for SQ; it is insignificant based on Moran's I yet significant in the GAM for MRS; and it is absent for Cost. For inference on spatial trends, the strongest effects align across diagnostics, showing statistically significant spatial structure where both Moran's I and the GAM indicate nonrandom geographic patterning. Carbon is valued most along the coast and in the central–northeast portion of the state, and less in the southwest. Jobs, SQ, and MRS display broadly similar patterns, with higher values inland—especially along the south–central–east corridor—and lower values in the northwest quadrant.

[INSERT FIGURE 5 HERE]

The spatial analysis of attribute-level coefficients supports the spatial-structure component of H2 and is consistent with the driver component of H1. Moran's I indicates that clustering in some attribute effects reflects systematic geographic structure rather than noise, while the spatial GAMs recover coherent preference gradients across Oregon for Jobs, Carbon, SQ, and MRS. Together, these diagnostics match H2's prediction that nearby residents exhibit more similar attribute preferences than distant residents and that preferences vary smoothly over space—most clearly for Jobs and Carbon, and more weakly for SQ and MRS - implying that preferences are spatially patterned rather than uniform.

In relation to H1, which posits that preference segments are shaped by attitudes and proximity to the coast, our results indicate that both segment membership and spatial attribute surfaces vary systematically with coastal distance. We find a clear coast-interior contrast: coastal communities place higher value on blue-carbon attributes, whereas in-

land communities place more weight on jobs and are more status-quo oriented. Consistent with this pattern, the GAM class-probability smooths show that classes with more environmentalist worldviews are more likely to be located closer to the coast, reinforcing a distance-decay relationship in both attitudes and blue-carbon preferences. In relation to H3, we reject the claim that marine reserve size (MRS) and Jobs exhibit stronger spatial structure and distance decay than carbon. We do not observe greater spatial dependence for MRS and Jobs relative to carbon.

6 Discussion and conclusions

This study evaluated Oregon residents' preferences for changes in marine-reserve policy in relation to marine reserve size, blue-carbon capture, and coastal jobs using a latent class model (LCM) combined with complementary spatial diagnostics (Global/Local Moran's I and bivariate spatial GAMs). The goal was to investigate deeper the heterogeneity in preferences and assess whether proximity to the coast and underlying attitudes predict preference segmentation and if preferences and segment membership are spatially clustered rather than randomly distributed taking into account the type of environmental good.

Overall, the LCM results indicate substantial heterogeneity in support for Oregon's MRS policy changes, with five well-differentiated segments and a clear majority that disfavors the status quo. The three largest segments - about three-quarters of respondents - systematically reject the status quo and value at least two program attributes, while the remaining segments place greater weight on avoiding change. The groups that support changes have more pro-environmental attitudes and more progressive worldviews. One considerable pro-change group is more likely to be located near the coast. The smaller change-averse group is more likely to be located farther from the coast. This provides a first approximation that pro-environmental attitudes/worldviews and coastal proximity are important elements that influence support to Marine Reserve policies (Ruiz-Frau et al., 2019; McAlpine et al., 2025; Sagebiel et al., 2016; Hoyos et al., 2015; Faccioli et al., 2020). Some segments reveal spatial structure, indicating that people living closer tend to reveal similar probabilities of belonging to a segment, forming geographic clusters of segment membership across the state. Also, some MRS attributes reveal spatial structure on its preferences, indicating that people value MR attributes more similarly when located nearer to one another. The spatial patterns detected corroborate a clear distinction between coastal and inland regions preferences. Regions closer to the coast (west) tend to value blue carbon more highly and are more likely to belong to groups that are less conservative and exhibit more pro-environmental attitudes. Inland clusters tend to place greater value on the status quo (avoiding changes) and on job generation through marine reserves.

This shows the importance of integrating spatial elements when analyzing stated choices preferences. As stated by works as Johnston Ramachandran (2014), Czajkowski et al. (2017), Campbell et al. (2009), and De Valck Rolfe (2018) this work corroborates with the view that those elements improve the estimates from choice models - LCM in our case - as confirmed by the bivariate spatial GAM diagnostics, where thin-plate regression splines captures smooth geographic variations and improve preferences estimates beyond the mean (non-spatial) logit es-timates (Wood, 2003; Wood, 2017).

In relation to use, non-use, global and non-global benefit implications, the spatial preferences story is not as straightforward as it may seem. The blue carbon attribute, which is a good that accrues global benefits, shows stronger preferences concentrated near the MRS, while Jobs, which is an attribute that accrues more local benefits, shows stronger preferences concentrated inland. Here, it is important to state that blue carbon preferences display spatial structure, in the sense that people approve it more in regions located nearer to the coastal ecosystems that capture it. Nevertheless, as the literature suggests (Faccioli et al., 2020; Hoyos et al., 2015; Eusse-Villa et al., 2021), the evaluation of carbon-related services is also mediated by pro-environmental attitudes and worldviews. In our case, there is an overlap between the cluster with a higher probability of belonging to Class 2 (which is characterized by more pro-environmental worldviews, stronger climate-change concern, greater eco-centrism and less conservative orientations) and the cluster of high valuation/marginal utility for the blue carbon attribute. This supports the view that preferences for environmental services accruing diffuse/global benefits can still be locally clustered closer to where the service is generated when the underlying attitudes and worldviews that shape valuation are also spatially concentrated in nearby the good that generates the ecosystem service.

Methodologically, the incorporation of GAM proved to be very valuable, reinforcing some Global Moran's I spatial autocorrelation results, adding robustness to the spatial-pattern findings, and providing an inferential approach that allows us to map and better understand what may be driving the spatial patterns. When an element is supported by both Moran's and GAM results, it reinforces the evidence of spatial structure, giving us more certainty in making spatial claims. When a result is not confirmed by Moran's test but is evidenced by GAM, it can provide insight that a spatial pattern may exist but varies more gradually or locally, not globally clustered (as the marginal utility for MRS attribute). Conversely, a spatial autocorrelation can be detected by Moran's test, but GAM can reveal that the pattern is too wiggly/complex and therefore unreliable (as in the case of Class 5). Therefore, this paper suggests that combining these approaches provides greater robustness for affirming evidence of spatial structure across distinct dimensions of preferences.

Policy wise, this work provides evidence to support that tailoring policy communication by place - according to the probability of belonging to a preference segment and/or

liking–disliking specific attributes - is a valuable strategy to consider, since welfare effects of the policy are distributed unevenly in space. Given that, instead of spreading outreach resources evenly with a uniform message, targeting communication in each region according to its dominant preference characteristics can increase support for MRS policy enhancement. The results presented here can, therefore, inform the design of place-specific messages, because we can better identify which policy elements, worldviews, and attitudes are more strongly supported in each region. Another point is that about one-quarter of the sample is averse to changes and supports the status quo, so identifying the regions where this segment is more prevalent helps direct communication and complementary incentives toward reducing perceived risk and increasing trust in MR policies in those regions.

The sample size in this research can be considered a limitation. Although our collected observations reflect the population distribution across space, a similar study with more observations could improve internal (and potentially external) validity and increase statistical power for detecting spatial structure. Also, instead of relying on ZIP-code information, future research could collect more detailed respondent location data (e.g., finer-scale geocodes), allowing spatial variation to be analyzed at a more granular coordinate level and reduce error measurement. We use ZIP codes and ZIP-code boundaries and jitter the observations in the spatial analysis to increase spatial variation, because many respondents share the same ZIP code. Future work should also replicate this analysis in other regions with different cultures and stages of marine-reserve's policy development, to better understand how spatial patterns in marine-reserve preferences vary with geography, climate context, and institutions.

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Figures and Tables

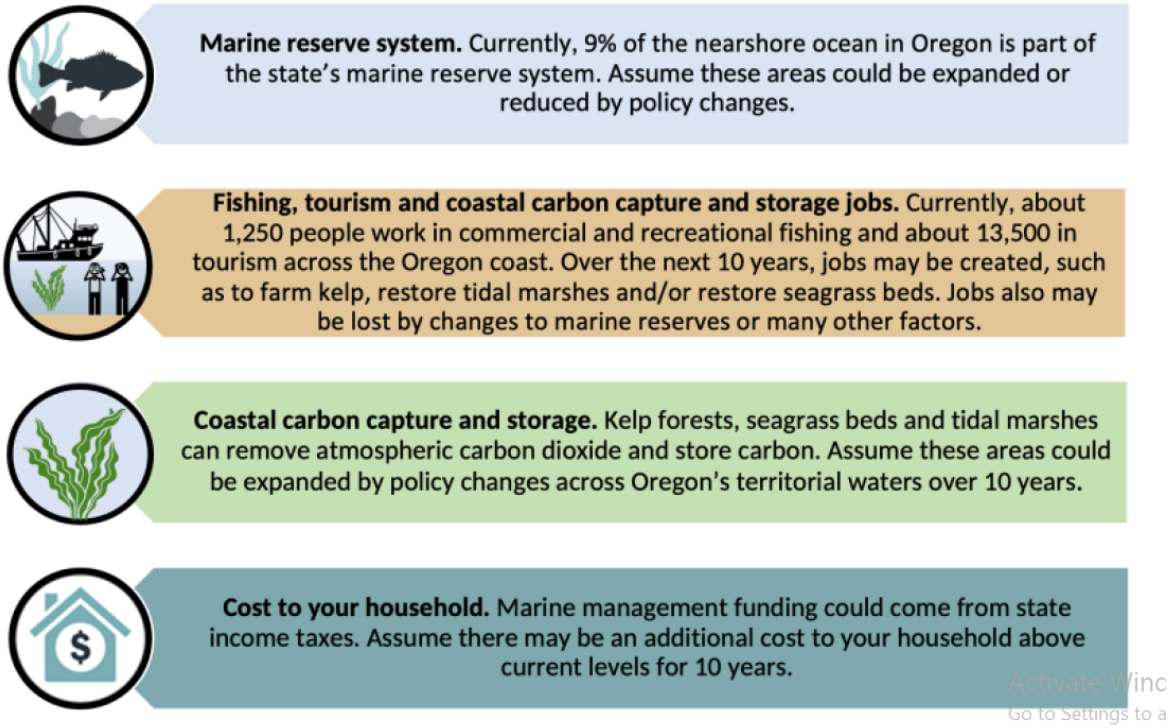
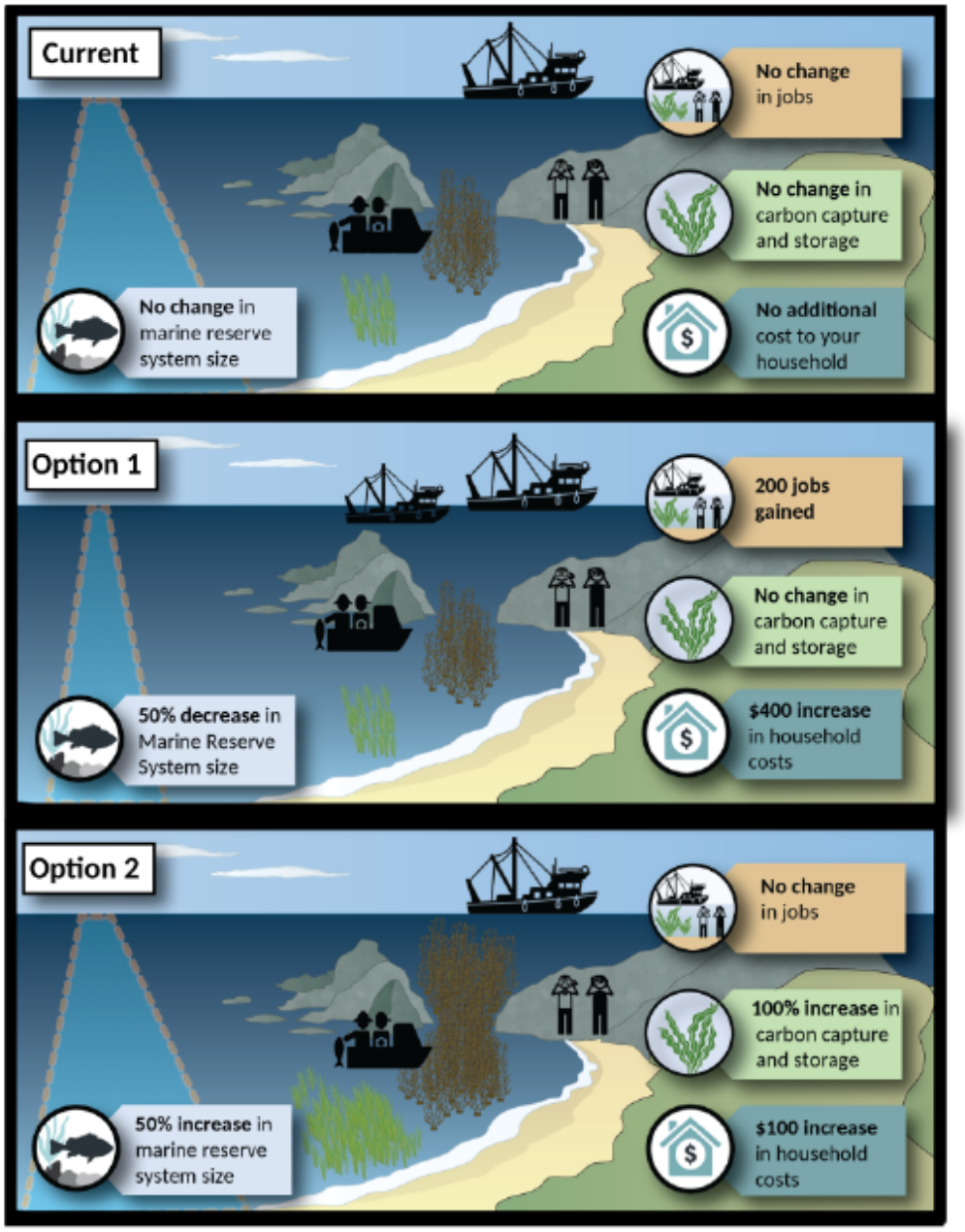


Figure 1: Panel A: Description of the marine-reserve attributes shown to respondents.



- ◯ Current situation (no change)
- ◯ Option 1
- ◯ Option 2

How certain are you that the choice you made in this comparison reflects your preferences?

0 = Not at all certain; 100 = Completely certain

0 10 20 30 40 50 60 70 80 90 100

Figure 1: Panel B: Description of the choice tasks shown to respondents.

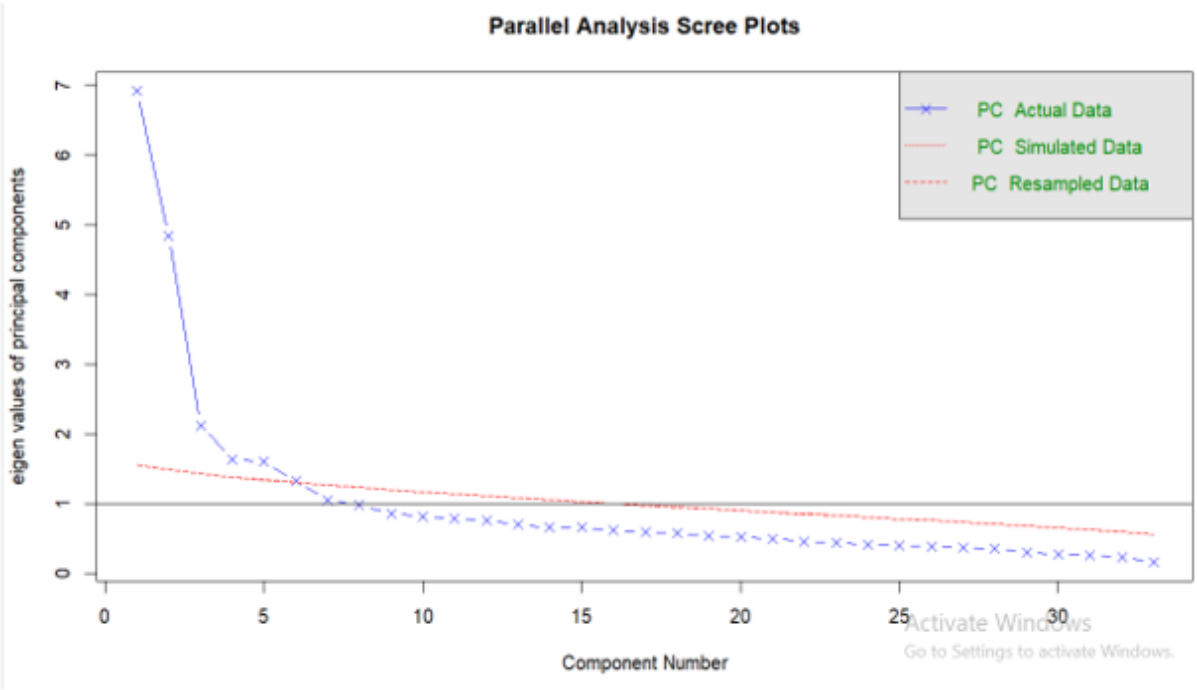


Figure 2: Parallel Analysis to define number of factors.

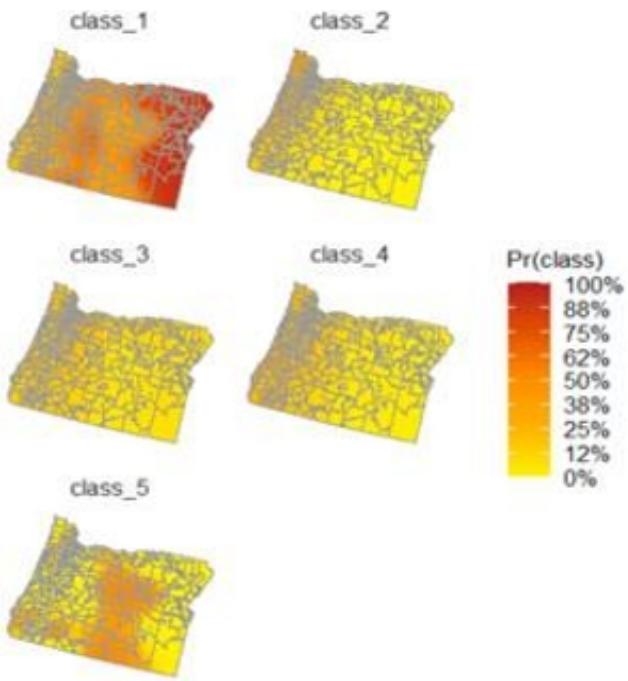


Figure 3: Spatial probability surfaces of latent-class membership from a spatial GAM with 2-D smooth

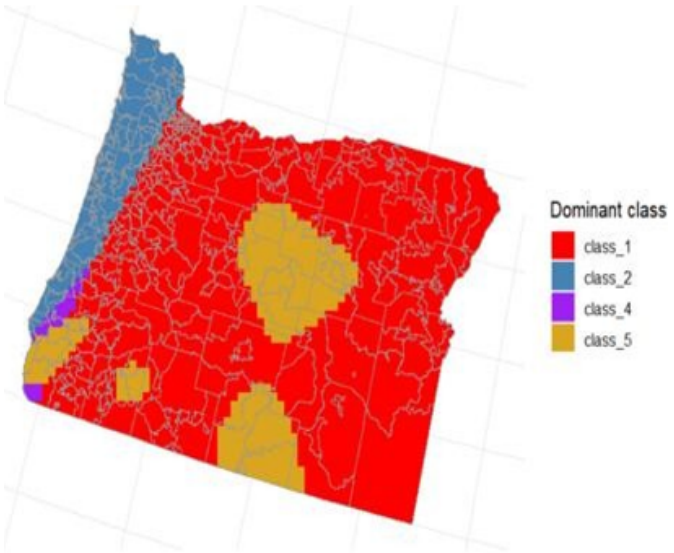


Figure 4: Argmax of Predicted Class-Membership Probabilities from Spatial GAM

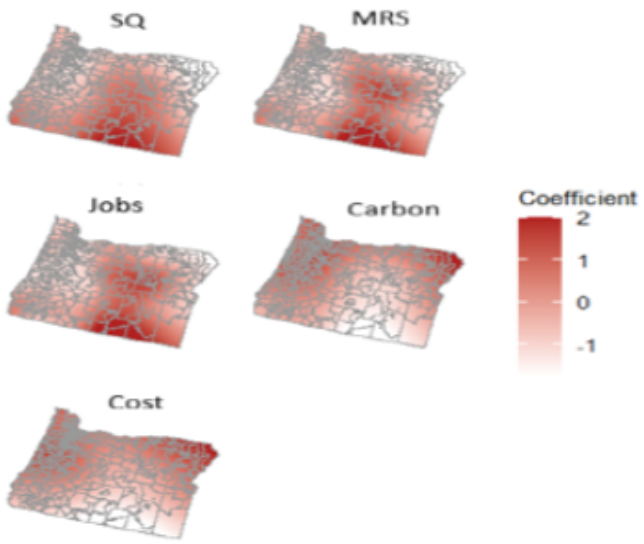


Figure 5: Spatial surfaces of LCM utility coefficients from a spatial GAM with 2-D smooth

Table 1: Choice experiment attributes (and corresponding variable names)

- **Size of Marine Reserve System** (*MRS*)
  - No change in size of marine reserve system (status quo).
  - 50% decrease in the size of the reserves.
  - 50% increase in the size of the reserves.
- **Marine Reserve System's Carbon-Capture Capacity** (*Carbon*)
  - No change in the reserves' carbon-capture capacity (status quo).
  - 25% increase in the reserves' carbon-capture capacity.
  - 50% increase in the reserves' carbon-capture capacity.
  - 100% increase in the reserves' carbon-capture capacity.
- **Oregon's Coast Net Jobs** (*Jobs*)
  - No change in number of jobs (status quo).
  - 200 jobs lost.
  - 200 new jobs created.
- **Annual Household Cost of Marine Reserve System** (*Cost*)
  - $0 (status quo)
  - $25
  - $50
  - $100
  - $200
  - $400
  - $500

Note: Variable names are provided in parentheses (in italics).

Table 2: Socio-demographic variables, sample percentages for the "=1" category, and census estimates.

Variable	**Description**	**% of Sample**[a]	**Census Estimate**[b]
gender[c]	=1 if male, 0 otherwise	61.2% (56.7, 65.7)	49.7%
senior	=1 if age is 55+ years, 0 otherwise	38.3% (33.8, 42.8)	31.5%
midage	=1 if age 35–54, 0 otherwise	40.4% (35.8, 45.0)	40.6%
white	=1 if identifies as White, 0 otherwise	89.3% (86.4, 92.2)	73.9%
latino	=1 if Hispanic/Latino, 0 otherwise	7.48% (5.02, 9.93)	14.9%
black	=1 if Black/African-American, 0 otherwise	2.49% (1.03, 3.94)	2.10%
native	=1 if Amer. Indian/Alaska Native, 0 otherwise	2.72% (1.20, 4.23)	1.10%
asian	=1 if Asian, 0 otherwise	4.31% (2.41, 6.20)	4.6%
pacific	=1 if Nat. Hawaiian/Pac. Isl., 0 otherwise	0.68%(–0.09, 1.45)	0.5%
other	=1 if Other race, 0 otherwise	3.63% (1.88, 5.37)	2.9%
college[d]	=1 if some college/Associate, 0 otherwise	36.51%(32.0, 41.0)	32.3%
bachdeg[d]	=1 if Bachelor's degree, 0 otherwise	29.3% (25.0, 33.5)	23.5%
graddeg[d]	=1 if Graduate degree, 0 otherwise	16.3% (12.9, 19.7)	14.2%
highinc	=1 if income $250k, 0 otherwise	3.40% (1.71, 5.09)	11.8%
himidinc	=1 if $100k income <$250k, 0 otherwise	28.1% (23.9, 32.3)	27.8%
midinc	=1 if $50k income <$100k, 0 otherwise	42.6% (38.0, 47.2)	29.5%
workft	=1 if working full-time, 0 otherwise	47.2% (42.5, 51.8)	60.0%
workpt	=1 if working part-time, 0 otherwise	10.2% (7.37, 13.0)	—
homemaker	=1 if homemaker, 0 otherwise	2.27% (0.88, 3.66)	37.4%
student	=1 if student, 0 otherwise	1.81% (0.56, 3.05)	—
retired	=1 if retired, 0 otherwise	19.0% (15.4, 22.7)	—
urban	=1 if urban/peri-urban area, 0 otherwise	27.9% (23.7, 32.1)	67.0%

Notes:

a 95% confidence intervals in parenthesis.
b Estimates obtained from the U.S. Census Bureau, American Community Survey, 2023, one year data profiles.
`https://data.census.gov/table/ACSDP1Y2023.DP05?g=040XX00US41` and
`https://data.census.gov/profile/Oregon?g=040XX00US41#education`.
c Respondents were able to identify as "other" or "prefer not to say". However, none chose either of these two personal identifiers.
d Population 25 years and older.
e Census estimate for `highinc` is $200,000 or greater annual household income, and for `himidinc` is at least $100,000 and no greater than $200,000.
f Population over 16 years of age employed in civilian labor force and armed services.
g Population over 16 years of age not in labor force.
h Obtained from
`https://www.ohsu.edu/oregon-office-of-rural-health/about-rural-and-frontier-data`.

Table 3: Promax Rotated – Factor Solution Applying EFA on the Whole Sample (n=441)

Scale items	**Climate Concern**	**Anthropocentrism**	**Eco-centrists**	**Conservatives/Risk Averse**	**Social Justice/Equalitarians**	**Uniqueness**
Clim_1	0.93					0.17
Clim_2	0.89					0.22
Clim_5	0.76					0.36
Clim_6	-0.72					0.41
TR_1		0.66				0.58
TR_2		0.72				0.54
TR_3		0.66				0.55
TR_4		0.77				0.37
TR_5		0.63				0.44
Rel_1			0.76			0.44
Rel_2			0.81			0.36
Rel_3			0.71			0.44
Rel_5			0.63			0.60
Pol_2					0.86	0.30
Pol_3					0.52	0.70
Pol_4					0.49	0.72
Schw_4				0.64		0.59
Schw_5				0.79		0.44
Schw_9				0.61		0.60
Variance explained	0.15	0.13	0.12	0.07	0.07	

Note: TLI = 0.929 (>0.9), RMSEA = 0.055 (<0.07), RMSR = 0.03 (<0.05). Cut-off values greater than 0.30.

Table 4: Latent Class Model fit from 1 to 10 classes specifications

# of Classes	1	2	3	4	5	6	7	8	9	10
AIC3	2956.4	2616.1	2596.8	2589.72	2574.7	2586.6	2592.2	2600.5	2597.6	2604.8
BIC	2970.9	2665.5	2681.2	2708.95	2728.8	2775.6	2816.1	2859.3	2891.3	2933.4
AIC	2951.4	2599.1	2567.8	2548.72	2521.7	2521.6	2515.2	2511.5	2496.6	2491.8
LL	-1470.7	-1282.5	-1254.9	-1233.4	-1207.8	-1195.8	-1180.6	-1166.7	-1147.3	-1132.9

Table 5: Estimates from Latent Class Model with 5 classes

	LCM – 5 classes				
	Class 1	Class 2	Class 3	Class 4	Class 5
Class names	Jobs/Carbon pragmatists	Size/Carbon environmentalists	Ambitious Expansionists	Status-quo job idealists	Strong status-quo keepers
Mg. Effects					
SQ	-0.291***	-0.272***	-0.271***	0.298***	1.625**
	(0.056)	(0.082)	(0.089)	(0.060)	(0.652)
MRS	0	0.003***	0.012***	0.001	0.055***
	(0.001)	(0.001)	(0.003)	(0.001)	(0.021)
Jobs	0.0005***	0.0003	0.001***	0.0005**	0.011***
	(0.0001)	(0.0002)	(0.0004)	(0.0002)	(0.004)
Carbon	0.002***	0.004***	0.007***	-0.004**	-0.017
	(0.0005)	(0.0009)	(0.0019)	(0.002)	(0.015)
Cost	-0.0005***	-0.0001	-0.003***	-0.0004	-0.004*
	(0.0001)	(0.0002)	(0.001)	(0.0003)	(0.002)
Covariates[a]					
Climate concern	-0.426*	2.357***	0.093	-0.763***	-1.261***
	(0.255)	(0.729)	(0.313)	(0.266)	(0.361)
Anthropocentrist	-0.112	0.357	0.077	0.764***	-1.086***
	(0.199)	(0.240)	(0.208)	(0.196)	(0.352)
Eco-centrist	0.014	0.569**	0.041	-0.329*	-0.294
	(0.167)	(0.274)	(0.211)	(0.190)	(0.288)
Conservative	0.015	-0.441*	0.586***	-0.317	1.329**
	(0.196)	(0.234)	(0.206)	(0.205)	(0.563)
Equalitarian	-0.257	-0.322	0.144	-0.631***	1.066*
	(0.221)	(0.303)	(0.270)	(0.227)	(0.648)
Travel Time	0.008**	-0.017**	0.001	-0.006	0.014***
	(0.003)	(0.008)	(0.005)	(0.004)	(0.004)
Class probability (π)	0.332***	0.217***	0.208***	0.186***	0.057***
	(0.059)	(0.048)	(0.047)	(0.024)	(0.014)
Log-likelihood	-1207.843				
AIC3	2574.686				
#parameters	53				

Note: Standard errors in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.
[a] Covariates are effects coded.

Table 6: Moran's I statistics for class probability scores

Class	I	$E[I]$	$\mathrm{Var}[I]$	z	p	p_{perm}
1	0.133	-0.00274	0.000531	5.89	1.88E-09	0.001
2	0.0628	-0.00274	0.000529	2.85	0.00217	0.006
3	0.00695	-0.00274	0.000529	0.422	0.337	0.307
4	0.0196	-0.00274	0.000528	0.973	0.165	0.173
5	0.0605	-0.00274	0.000511	2.8	0.00255	0.012

Note: $N = 366$ for this test, where 5 observations reported post offices numbers instead of zip codes, therefore not detected in zip-code shapefile.

Table 7: Spatial GAM diagnostics for class-membership probabilities: smooth $s(x, y)$ significance and incremental fit vs. null

	GAM diagnostics for $s(u, v)$					Incremental fit			
	edf	F	p	dev_{expl}	R^2_{adj}	df_{diff}	dev_{diff}	F_{inc}	p_{inc}
Class 1	2	17.386	1e-08	0.102	0.118	2.002	25.088	22.927	4.1e-10
Class 2	2.044	9.806	0.00006	0.091	0.060	2.132	21.31	16.464	6.6e-08
Class 3	4.8	1.22	0.253	0.035	0.019	8.569	6.387	1.508	0.147
Class 4	2.051	2.493	0.082	0.020	0.011	2.153	5.779	3.313	0.034
Class 5	36.458	1.268	0.124	0.430	0.295	46.957	56.825	1.95	0.00045

Table 8: Moran's I statistics for attribute estimates

Attributes	I	$E[I]$	$\mathrm{Var}[I]$	z	p	p_{perm}
SQ	0.0298	-0.00274	0.000522	1.42	0.0773	0.087
MRS	0.0238	-0.00274	0.000513	1.17	0.1210	0.140
Job	0.0448	-0.00274	0.000513	2.10	0.0179	0.038
Carbon	0.0385	-0.00274	0.000525	1.80	0.0360	0.038
Cost	-0.00666	-0.00274	0.000529	-0.171	0.5680	0.562

Table 9: Spatial GAM diagnostics for attributes: smooth $s(x, y)$ significance and incremental fit vs. null

	GAM diagnostics for $s(x, y)$					Incremental fit			
	edf	F	p	$\mathrm{dev}_{\mathrm{expl}}$	R^2_{adj}	df_{diff}	$\mathrm{dev}_{\mathrm{diff}}$	F_{inc}	p_{inc}
SQ	9.53	1.3540	0.17800	0.077	0.053	16.724	0.0110	1.782	0.030
MRS	23.05	1.8420	0.00482	0.203	0.149	34.940	0.1770	2.486	1.7E-05
Job	23.46	2.1300	0.00050	0.222	0.168	35.036	0.0050	2.778	1.2E-06
Carbon	8.752	1.7522	0.05300	0.081	0.058	14.776	0.0150	2.114	0.009
Cost	6.332	1.1579	0.31500	0.043	0.026	10.186	0.0004	1.599	0.104

Appendix 1. Survey Scale Items Used to Measure Latent Constructs

Scale	Items
Climate Urgency	
Clim_1	The human causes of climate change need to be addressed immediately, even if the cost of doing that is substantial.
Clim_2	Human impacts are causing changes at a planetary scale that are destabilizing the Earth's systems.
Clim_5	Hundreds of years from now a future scientist will see this period as a great extinction event, this one caused by climate change.
Clim_6	Climate change does not pose an immediate risk to the environment.
Teddy Roosevelt / Anthropocentrism	
TR_1	The main reason to protect the environment is to conserve fish and other natural resources for future human benefit.
TR_2	Nature is important mostly because of what it can contribute to human well-being.
TR_3	The main goal of natural resource management should be to provide a stable flow of resources to sustain jobs and local communities.
TR_4	Natural resource management should focus primarily on benefit to humans.
TR_5	Plants and animals exist primarily for the benefits of humans.
Relational Values	
Rel_1	I have strong feelings about nature (including all plants, animals, the land, etc.). These views are part of who I am and how I live my life.
Rel_2	Plants and animals, as part of the interdependent web of life, are like *'kin'* or family to me.
Rel_3	I often think of some wild places whose fate I care about and strive to protect, even though I may never see them myself.
Rel_5	I think of animals not as groups, such as a herd of deer, but as individual animals, each with its own personality.
Political Values	
Pol_2	Equality – everyone has the same chance to get ahead in life.
Pol_3	Economic security – everyone has a steady job and decent income.

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Scale	**Items**
Pol_4	Morality – people live according to the rules of decent behavior.
Schwartz Values	
Schw_4	They believe that people should do what they're told and follow rules at all times, even when no one is watching them.
Schw_5	They want to always behave properly, avoid doing anything people would say is wrong.
Schw_9	They value security and avoid anything that might endanger their safety.

Appendix 2. Polychoric correlations between statements

	Clim_1	Clim_2	Clim_5	Clim_6	TR_1	TR_2	TR_3	TR_4	TR_5	Rel_1	Rel_2	Rel_3	Rel_5	Pol_2	Pol_3	Pol_4	Schw_4	Schw_5	Schw_9
Clim_1	1																		
Clim_2	0.85	1																	
Clim_5	0.76	0.75	1																
Clim_6	-0.74	-0.72	-0.64	1															
TR_1	-0.16	-0.17	-0.06	0.26	1														
TR_2	-0.08	-0.10	-0.12	0.22	0.53	1													
TR_3	-0.17	-0.16	-0.10	0.32	0.51	0.53	1												
TR_4	-0.32	-0.34	-0.20	0.49	0.48	0.48	0.51	1											
TR_5	-0.41	-0.44	-0.32	0.55	0.42	0.42	0.42	0.72	1										
Rel_1	0.46	0.42	0.43	-0.31	0.02	0.02	0.03	-0.22	-0.29	1									
Rel_2	0.43	0.43	0.42	-0.30	-0.02	-0.02	-0.07	-0.27	-0.36	0.62	1								
Rel_3	0.50	0.49	0.45	-0.36	-0.01	-0.05	-0.04	-0.24	-0.29	0.64	0.59	1							
Rel_5	0.39	0.35	0.32	-0.15	0.01	0.00	-0.04	-0.17	-0.26	0.43	0.57	0.48	1						
Pol_2	0.30	0.27	0.27	-0.27	0.08	0.12	0.07	-0.10	-0.15	0.29	0.23	0.24	0.19	1					
Pol_3	0.23	0.20	0.20	-0.15	0.04	0.00	0.12	-0.04	-0.15	0.23	0.22	0.20	0.15	0.60	1				
Pol_4	-0.02	0.01	0.01	-0.03	0.13	0.16	0.16	0.04	0.02	0.14	0.16	0.07	0.07	0.51	0.38	1			
Schw_4	-0.06	-0.02	-0.01	0.13	0.22	0.23	0.22	0.22	0.20	0.00	-0.04	-0.02	0.03	0.17	0.09	0.27	1		
Schw_5	-0.05	-0.03	0.00	0.18	0.17	0.14	0.25	0.22	0.18	0.08	0.04	0.05	0.07	0.13	0.14	0.28	0.53	1	
Schw_9	0.02	0.05	0.09	0.07	0.22	0.22	0.27	0.16	0.16	0.04	0.09	0.03	0.18	0.18	0.23	0.20	0.46	0.50	1

Appendix 3. Factor correlations

	Climate Concern	Conservative	Anthropocentrism	Equalitarian	Ecocentrism
Climate Concern	1				
Conservative	-0.38	1			
Anthropocentrism	0.55	-0.19	1		
Equalitarian	-0.12	0.43	0.04	1	
Ecocentrism	0.20	0.05	0.32	0.35	1

Appendix 4. Utility parameters from Latent Class Model with 5 classes

	LCM – 5 classes				
	Class 1	Class 2	Class 3	Class 4	Class 5
Class names	Jobs/Carbon pragmatists	Size/Carbon environmentalists	Ambitious Expansionists	Status-quo job idealists	Strong status-quo keepers
Attributes					
SQ	-1.308***	-1.226***	-1.220***	1.341***	7.311**
	(0.251)	(0.370)	(0.401)	(0.270)	(2.936)
MRS	-0.0001	0.015***	0.053***	0.006	0.246**
	(0.002)	(0.004)	(0.014)	(0.002)	(0.096)
Jobs	0.002***	0.001*	0.006***	0.002***	0.043***
	(0.0005)	(0.001)	(0.002)	(0.001)	(0.016)
Carbon	0.008***	0.0180***	0.030***	-0.016**	-0.075
	(0.002)	(0.0040)	(0.008)	(0.007)	(0.069)
Cost	-0.002***	-0.0005	-0.014***	-0.002*	-0.020*
	(0.0005)	(0.0008)	(0.0037)	(0.001)	(0.011)

Note: Standard errors in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Appendix 5. Climate Urgency, Relational, and Recreational Variables.

Variable Name	**Description**	**Sample Mean[a]**
Climate Urgency		
Clim_1	The human causes of climate change need to be addressed immediately, even if the cost of doing that is substantial.	2.58 (0.74)
Clim_2	Human impacts are causing changes at a planetary scale that are destabilizing the Earth's ecosystems.	2.59 (0.74)
Clim_3	The suggestion that climate change will cause large-scale ecosystem changes is overblown.	1.76 (0.86)
Clim_4	Hundreds of years from now a future scientist will see this period as a great extinction event, this one caused by climate change.	2.35 (0.85)
Clim_5	Climate change does not pose an immediate risk to the environment.	1.62 (0.82)
Relational		
Rel_1	I have strong feelings about nature (including all plants, animals, the land, etc.). These views are a part of who I am and how I live my life.	2.67 (0.59)
Rel_2	Plants and animals, as part of the interdependent web of life, are like 'kin' or family to me.	2.48 (0.73)
Rel_3	I often think of some wild places whose fate I care about and strive to protect, even though I may never see them myself.	2.58 (0.68)
Rel_4	People have a responsibility to account for their impacts on the environment because those impacts can harm other people.	2.85 (0.47)
Rel_5	I think of animals not as groups, such as a herd of deer, but as individual animals, each with its own personality.	2.29 (0.77)
Rel_6	People should take only what they need from nature and leave the rest behind for nonhuman beings.	2.83 (0.45)

(continued)

Variable Name	**Description**	**Sample Mean[a]**
Recreation		
Rec_1	Ocean-oriented sightseeing, photography, and/or wildlife viewing (birds, whales, etc.).	2.39 (0.59)
Rec_2	Beach walking, running, shell collecting and/or rock-hounding.	2.60 (0.52)
Rec_3	Exploring tide pools.	2.20 (0.67)
Rec_4	Ocean swimming, kayaking, surfing, boogie boarding, snorkeling, and/or diving.	1.64 (0.71)
Rec_5	Ocean recreational fishing (including charter) and/or collecting shellfish (e.g., crabs or clams).	1.75 (0.76)
Rec_6	Other ocean recreation.	1.99 (0.71)

[a] For Climate Change and Relational scores range from 1 (respondent strongly disagrees with the statement) to 7 (respondent strongly agrees). For Recreation scores range from 1 (respondent never participates) to 3 (respondent participates often, more than 10 times in the lst five years). Standard deviations are in parentheses.